The documentation and process conversion measures necessary to comply with this document shall be completed by 29 September 2026.

INCH-POUND

MIL-PRF-19500/117T
w/AMENDMENT 3
29 May 2026
SUPERSEDING
MIL-PRF-19500/117T
w/AMENDMENT 2
19 March 2025

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, SILICON, VOLTAGE REGULATOR,
TYPES 1N962 THROUGH 1N992, 5, 2, AND 1 PERCENT VOLTAGE TOLERANCE, ENCAPSULATED (AXIAL LEAD THROUGH-HOLE AND SURFACE MOUNT PACKAGES) AND UN-ENCAPSULATED (DIE),
QUALITY LEVELS JAN, JANTX, JANTXV, JANHC, AND JANKC

JANS level (see 6.6).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 500 milliwatt, silicon, voltage regulator diodes with voltage tolerances of 5, 2, and 1 percent. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two level of product assurance is provided for each unencapsulated device type.

1.2 Physical dimensions. The device package for the encapsulated device type are as follows: Axial leaded (DO-204AA, formerly DO-35) in accordance with figure 1, glass body surface mount (DO-213AA) in accordance with figure 2, UB in accordance with figure 3, UB2, and in accordance with figure 4. and UD in accordance with figure 5. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: A, B, and C version die in accordance with figure 6, D version die in accordance with figure 7, and E version die in accordance with figure 8.

1.3 Maximum ratings. Maximum ratings are as shown in maximum and primary test ratings (see 3.6.2) herein and as follows:

a. P_{TL} = 500 mW, (DO-204AA) at T_L = +50°C, L = .375 inch (9.53 mm); both ends of case or diode body to heat sink at L = .375 inch (9.53 mm). Derate I_Z to 0.0 mA dc at +175°C.

b. P_{TEC} = 500 mW (DO-213AA) at T_{EC} = +125°C, derate to 0 at +175°C.
-65°C ≤ T_J ≤ +175°C; -65°C ≤ T_{STG} ≤ +175°C.

c. $P_{T(PCB)}$ = 400 mW (DO-204AA), T_A = +55°C. Derate to 0 at +175°C.

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1.3 Maximum ratings. – continued.

d. $P_{SP(IS)}$ = 500 mW (UB, UB2, and UD) at $T_{SP(IS)}$ = +125°C. (Derate to 0 at +175°C). -65°C ≤ T_J ≤ +175°C; -65°C ≤ T_{STG} ≤ +175°C.

e. P_{TPCB} = 400 mW (UB, UB2, and UD) at T_A = +75°C. (Derate to 0 at 175°C.)

(1) See figures 9 through 13 for derating curves.

(2) T_A = +75°C (in still air) for both axial leaded and UR mounted horizontally on printed circuit board (PCB). The details for the PCB are as follows. Base material is FR4 .0625 inch (1.59 mm) thick; conductor layer is 1-layer 1-oz Cu. PCB pad size for UR devices is .067 inch (1.70 mm) x .105 inch (2.67 mm); PCB land sizes for axial leaded devices is .092 inch (2.34 mm) diameter. Strips connected to the lands and pads are .030 inch (0.762 mm) x 1 inch (25.4 mm) long. Axial leaded length L ≤ .187 inch (≤ 4.76 mm).

1.4 Primary electrical characteristics. Primary electrical characteristics are as shown in maximum and primary test ratings (see 3.6.2) herein and as follows:

a. 11 V dc ≤ V_Z ≤ 200 V dc.

b. 1N962B-1 through 1N992B-1 are ±5 percent voltage tolerance.

c. 1N962C-1 through 1N992C-1 are ±2 percent voltage tolerance.

d. 1N962D-1 through 1N992D-1 are ±1 percent voltage tolerance.

e. For thermal impedance curves see figures 14 through 20.

f. $R_{\theta JL}$ = 250°C/W maximum at L = .375 inch (9.53 mm) (DO-35).

g. $R_{\theta JEC}$ = 100°C/W maximum. Junction to end-caps (DO-213AA).

h. $R_{\theta JA(PCB)}$ = 300°C/W junction to ambient including PCB. $R_{\theta JA(PCB)}$ with a defined thermal resistance condition included is measured at I_Z = as defined in the electrical characteristics tolerance table herein.

i. $R_{\theta JSP(IS)}$ = 90°C/W (maximum) junction to solder pad (IS) (UB).

j. $R_{\theta JA(PCB)}$ = 250°C/W (maximum) junction to ambient (PCB) (UB).

k. $R_{\theta JSP(IS)}$ = 80°C/W (maximum) junction to solder pad (IS) (UD).

l. $R_{\theta JA(PCB)}$ = 228°C/W (maximum) junction to ambient (PCB) (UD).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level designators.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", and "JANTXV".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

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1.5.2 Device type. The designation system for the device types of semiconductor covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the semiconductor covered by this specification sheet are as follows: "962", "963", "964", "965", "966", "967", "968", "969", "970", "971", "972", "973", "974", "975", "976", "977", "978", "979", "980", "981", "982", "983", "984", "985", "986", "987", "988", "989", "990", "991", and "992".

1.5.3 Suffix symbols. The following suffix symbol are incorporated in the PIN as applicable.

1.5.3.1 First suffix symbol (voltage tolerance). The first suffix letter "B" indicates that the diodes are ±5 percent voltage tolerance. The first suffix letter "C" indicates that the diodes are ±2 percent voltage tolerance. The first suffix letter "D" indicates that the diodes are ±1 percent voltage tolerance.

1.5.3.2 Second suffix symbols (package designators). The following suffix symbols are incorporated in the PIN for this specification sheet:

-1	Indicates an axial through hole DO-35 metallurgical bonded double plug construction (see figure 1).
UR-1	Indicates a surface mount DO-213AA using a metallurgical bonded double plug construction (see figure 2)
UB	Indicates a 3 terminal, 4 pad surface mount package. The metal lid is connected to pad 4 (see figure 3).
UB2	Indicates a 3 terminal, 4 pad surface mount package. The metal lid is connected to pad 4, (see figure 4).
UD	Indicates a 2 terminal, 3 pad surface mount package (figure 5).

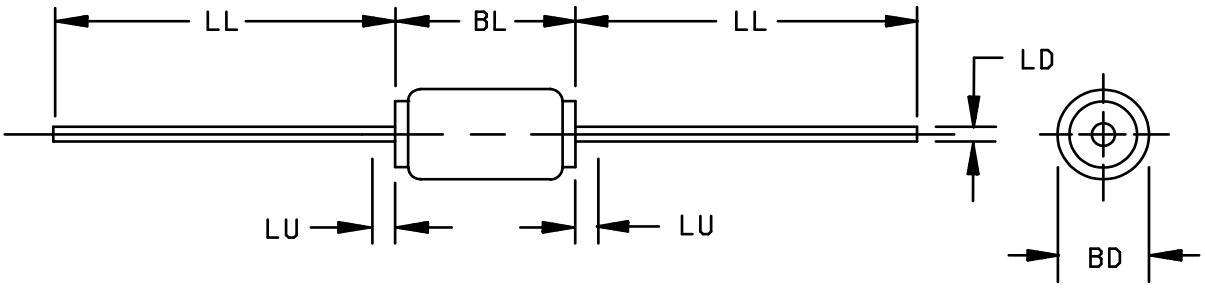
1.5.3.3 Internal circuit schematic designator suffix symbols (for UB and UB2 packages only). The suffix symbols that designate the internal circuit schematic (see schematics on figure 3) of the surface mount packages are as follows:

	No suffix Indicates a unidirectional, single diode circuit (UB package only).
CA	Indicates dual diode with a common anode pin (UB package only).
CC	Indicates a dual diode with a common cathode pin (UB package only).
D	Indicates a dual diode doubler (UB package only).
R	Indicates a reverse polarity configuration (UB2 package only).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A", "B", "C", "D" and "E".

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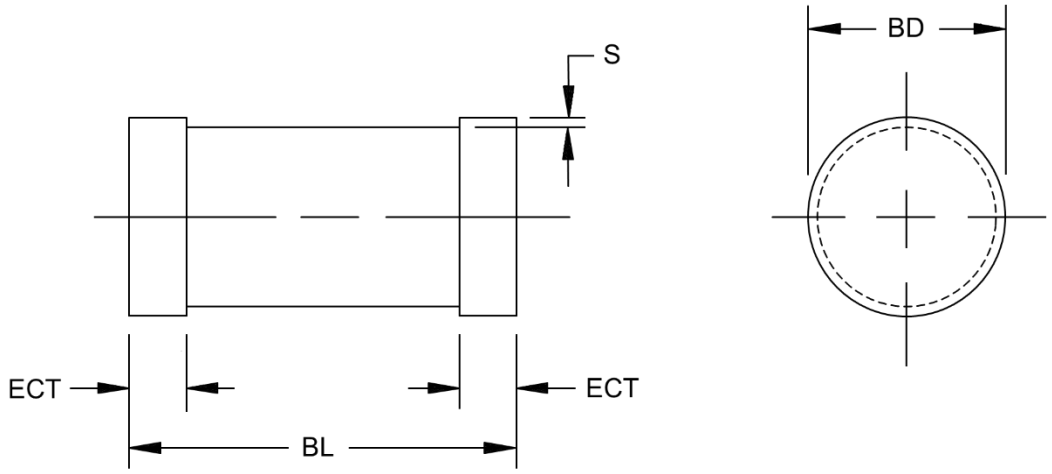


Symbol	Dimensions				Notes
	inch	inch	mm	mm	
	Min	Max	Min	Max	
BD	.055	.090	1.40	2.29	3
BL	.120	.200	3.05	5.08	3
LD	.018	.022	0.46	0.56	
LL	1.000	1.500	25.40	38.10	
LU		.050		1.27	4

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Package contour optional within dimension BD and length BL. Heat slugs, if any, shall be included within this but shall not be subject to minimum limit of LD.
4. The BL dimension shall include the entire body including slugs.
5. Within LL_1 lead diameter may vary to allow for flash, lead finish build-up, and minor irregularities other than heat slugs.

FIGURE 1. Physical dimensions for axial leaded DO-204AH (formerly DO-35) package.

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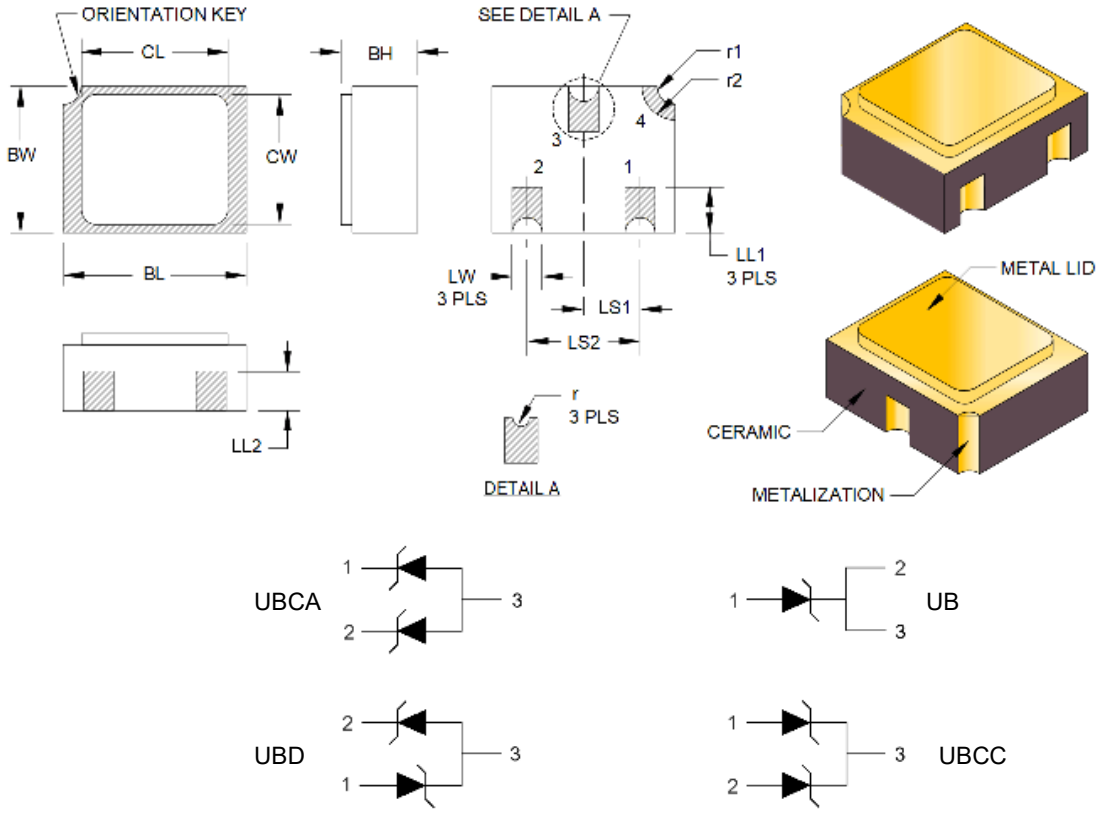


Symbol	Dimensions				Notes
	inch	inch	mm	mm	
	Min	Max	Min	Max	
BL	.130	.146	3.30	3.71	
BD	.063	.067	1.60	1.70	2, 3
ECT	.016	.022	0.41	0.55	
S	0		0		3

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension S is optional however, dimension BD (glass body) shall not exceed endcap diameter.

FIGURE 2. Physical dimensions for DO-213AA (UR) surface mount package.

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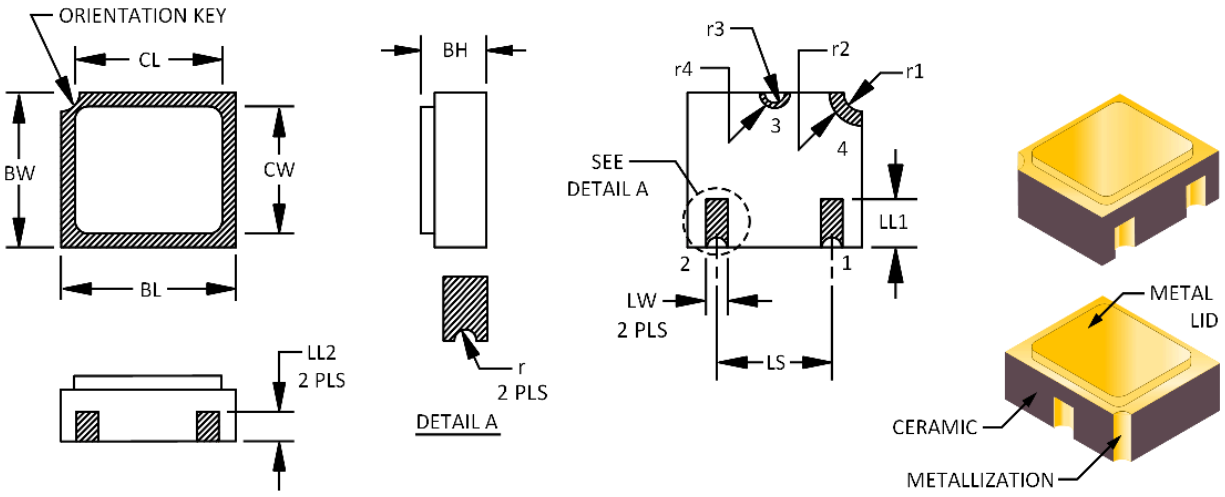


Symbol	Dimensions				Symbol	Dimensions			
	inch	inch	mm	mm		inch	inch	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.040	.056	1.02	1.42	LS1	.035	.039	0.89	0.99
BL	.115	.128	2.92	3.25	LS2	.071	.079	1.80	2.01
BW	.085	.108	2.16	2.74	LW	.016	.024	0.41	0.61
CL		.128		3.25	r		.008		0.20
CW		.108		2.74	r1		.012		0.31
LL1	.022	.038	0.56	0.97	r2		.022		0.56
LL2	.017	.035	0.43	0.89					

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = shielding, connected to the lid.
4. Dimensions are pre-solder dip.

FIGURE 3. Physical dimensions for 3 terminal UB surface mount package.

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POLARITY

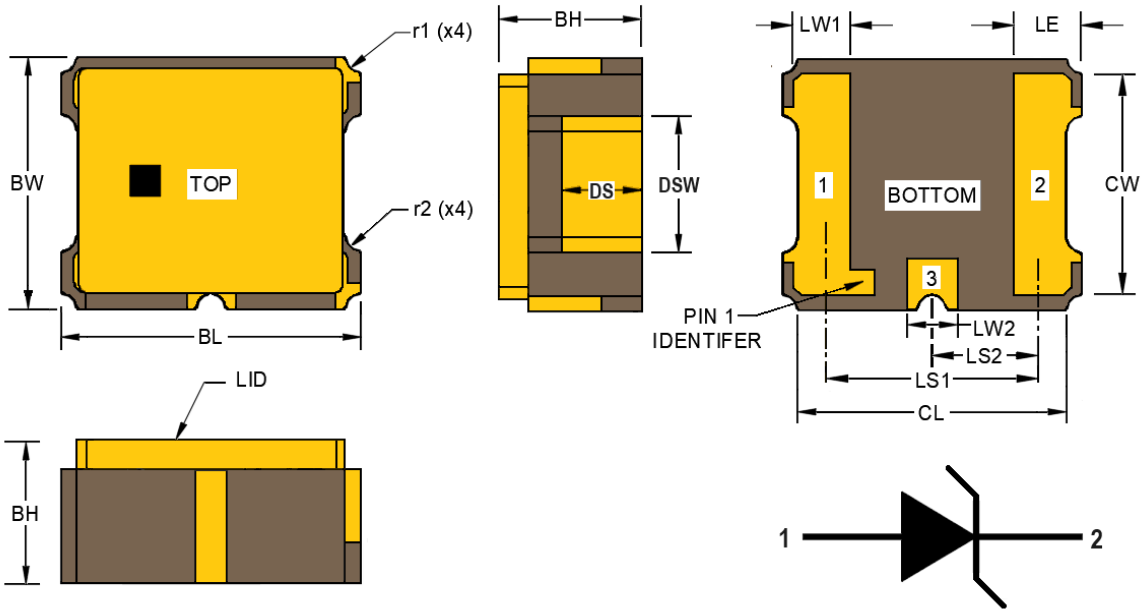


Symbol	Dimensions				Symbol	Dimensions			
	inch	inch	mm	mm		inch	inch	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.040	.056	1.02	1.42	LS	.071	.079	1.80	2.01
BL	.115	.128	2.92	3.25	LW	.016	.024	0.41	0.61
BW	.085	.108	2.16	2.74	r	.008 TYP		0.20 TYP	
CL		.128		3.25	r1	.012 TYP		0.30 TYP	
CW		.108		2.74	r2	.022 TYP		0.56 TYP	
LL1	.022	.038	0.56	0.96	r3	.008 TYP		0.20 TYP	
LL2	.017	.035	0.43	0.89	r4	.012 TYP		0.30 TYP	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = shielding, connected to the lid.
4. Dimensions are pre-solder dip.

FIGURE 4. Physical dimensions for 2 terminal UB2 surface mount package.

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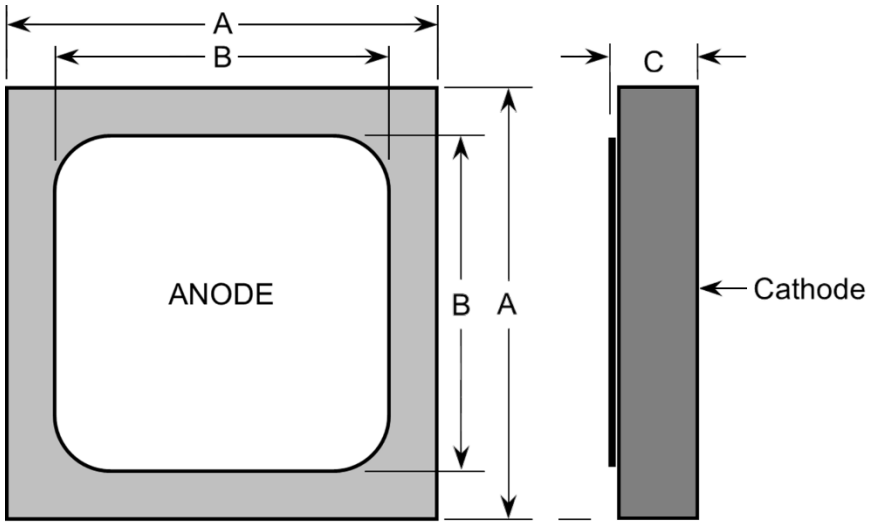


Symbol	inch Min	inch Max	mm Min	mm Max	Symbol	inch Min	inch Max	mm Min	mm Max
BH	0.078	0.092	1.979	2.329	LE	0.021	0.033	0.540	0.840
BL	0.112	0.128	2.850	3.250	LS1	0.085	0.093	2.160	2.360
BW	0.092	0.108	2.340	2.740	LS2	0.043	0.047	1.080	1.180
CL	0.103	0.113	2.620	2.880	LW1	0.019	0.027	0.490	0.690
CW	0.083	0.093	2.120	2.360	LW2	0.019	0.021	0.490	0.530
DS	0.028	0.034	0.720	0.880	r1		0.006		0.150
DSW	0.051	0.057	1.330	1.430	r2		0.006		0.150

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Pad 3 is connected to the lid braze ring for shielding.
3. Cathode marked with a square.
4. Pads 1 and 2 layout is equivalent to those for 1N962 through 1N992, and 1N962UR-1 through 1N992UR-1 (all voltage tolerance levels).
5. Diode outline shown in ISO third angle projection.

FIGURE 5. Physical dimensions of 2 terminal UD surface mount package.

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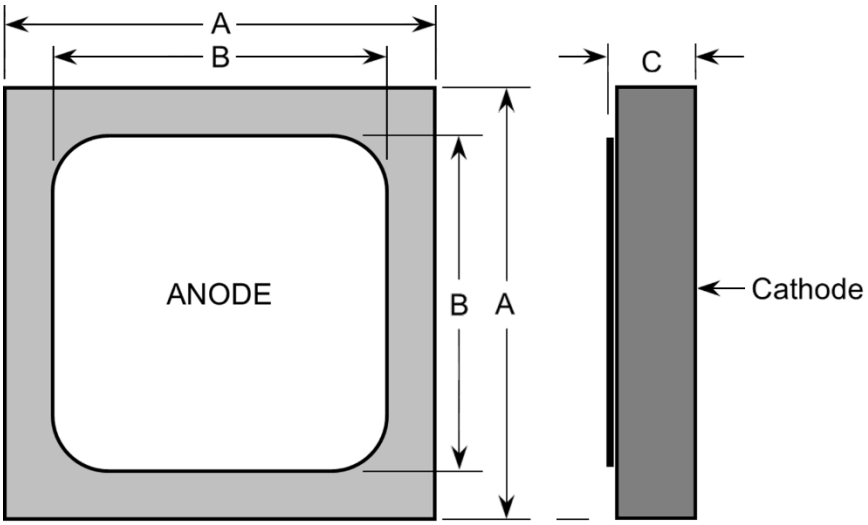


Letter	Dimensions											
	JANHCA				JANHCB				JANHCC, JANKCC			
	inch	inch	mm	mm	inch	inch	mm	mm	inch	inch	mm	mm
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
A	.021	.025	0.53	0.63	.024	.028	0.61	0.71	.019	.023	0.48	0.58
B	.013	.017	0.33	0.43	.017	.021	0.43	0.53	.013	.017	0.33	0.43
C	.008	.012	0.20	0.30	.008	.012	0.20	0.30	.008	.012	0.20	0.30

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Circuit layout data: For zener operation, cathode must be operated positive with respect to anode.
3. Element evaluation performed in a TO-5 package (see 6.5).
4. The metallization characteristics of the die.
   Top = (anode) - AL, minimum AL thickness =
   12,000 Å for JANHCA, 40,000 Å for JANHCB, and
   25,000 Å for JANHCC and KCC.
   Back: (cathode) - AU, minimum AU thickness =
   3,000 Å for JANHCA, and 5,000 Å for JANHCB, and
   4,000 Å for JANHCC and KCC.

FIGURE 6. Physical dimensions (JANHCA, JANHCB, JANHCC, and JANKCC die dimensions).

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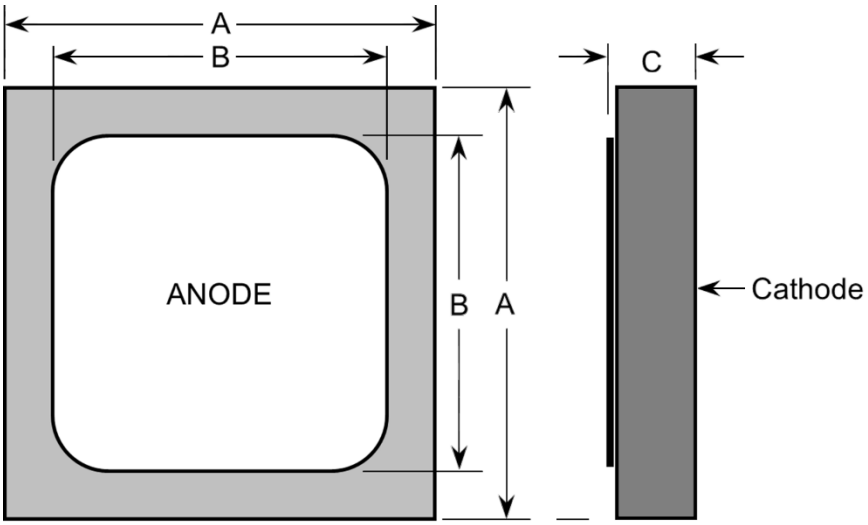


Letter	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.023	.027	0.584	0.686
B	.015	.019	0.381	0.483
C	.008	.012	0.203	0.305

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. The metallization characteristics of the die are as follows:
   Anode metallization: Al, thickness = 34,000 Å minimum;
   Cathode metallization: Ti/Au, thickness = Ti/Au (200 Å / 4,350 Å) nominal.

FIGURE 7. Physical dimensions (JANHCD, and JANKCD die dimensions).

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Letter	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.011	.015	0.28	0.38
B	.005	.009	0.13	0.23
C	.006	.010	0.15	0.25

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. The metallization characteristics of the die are as follows:
   Anode metallization: Al, thickness = 20,000 Å minimum.
   Cathode metallization: Au, thickness = 8,000 Å nominal.

FIGURE 8. Physical dimensions for JANHCE and JANKCE die (E version).

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2. APPLICABLE PUBLICATIONS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL−PRF−19500 − Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL−STD−750 − Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-19500, and as follows.

ATE	Automatic test equipment.
$R_{\theta JA}$	Thermal resistance, junction to ambient.
$R_{\theta JA(PCB)}$	Thermal resistance, junction to ambient, a defined PCB mounting.
$R_{\theta JEC}$	Thermal resistance, junction to end-cap.
$R_{\theta JL}$	Thermal resistance, junction to lead.
$R_{\theta JSP(IS)}$	Thermal resistance, junction to solder pad, infinite sink.
T_{EC}	Temperature of end-cap.

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3.4 Interface and physical dimensions. The interface and physical dimensions shall be specified in MIL-PRF-19500 and figures 1 and 2 (similar to DO-35 and DO-213AA), and figure 3 (UB), figure 4 (UB2), figure 5 (UD), figures 6, 7 and 8 (die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. All devices shall be metallurgically bonded double plug construction in accordance with the requirements of category I, II, or III (see MIL-PRF-19500).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500 and as specified herein. Manufacturer's identification and date code shall be marked on the devices. Initial container package marking shall be in accordance with MIL-PRF-19500. The prefixes JAN, JANTX, and JANTXV can be abbreviated as J, JX, and JV respectively. (Example: The PIN for an axial leaded device 1N962B-1 can be reduced to J962B1, JX962B1, or JV962B1.) No color coding shall be permitted for part identification.

3.5.1 Axial leaded devices. For axial leaded versions only, the polarity shall be indicated with a contrasting color band to denote the cathode end.

3.5.2 UR devices. For UR suffix devices only, all marking, except polarity (and serial number for JANS) may be omitted from the body. Polarity shall be identified with a cathode band as a minimum. Alternatively, a minimum of three evenly spaced contrasting color dots around the periphery of the cathode end may be used.

3.5.3 UB and UB2 devices. The 'UB' and UB2 version devices do not require polarity marking.

3.5.4 UD devices. The 'UD' devices shall mark the cathode with a square symbol on the lid.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I, table II, and table III.

3.6.1 Selection of tight tolerance devices. The C and D suffix devices shall be selected from JAN, JANTX, or JANTXV devices, which have successfully completed all applicable screening, and groups A, B, and C testing as 5 percent tolerance devices. All sublots of C and D suffix devices shall pass table I, subgroup 2, at tightened tolerances. Tighter tolerances for mounting clip temperature shall be maintained for reference purpose to establish correlation. For C and D tolerance levels, T_L = +25°C ±2°C at .375 inch (9.53 mm) from body or equivalent.

3.6.2 Maximum and primary test ratings. Maximum and primary test ratings for voltage regulator diodes are specified in table IV herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation of un-encapsulated die (see 4.6).

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC devices. JANHC and JANKC devices shall be qualified in accordance with appendix G of MIL-PRF-19500.

4.2.3 Construction verification. Cross sectional photos from three devices shall be submitted in the qualification report.

4.3 Screening (JAN, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2)
7a 7b	Not required Optional
9	Not required
11	I_{R1} and V_Z
12	See 4.3.3
(2) 13	ΔI_{R1} ≤ 100 percent of initial reading or 50 nA dc, whichever is greater. ΔV_Z ≤ ±2 percent initial reading. Subgroup 2 of table I herein.

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.
(2) PDA = 5 percent for screen 13 applies to ΔI_{R1} and ΔV_Z. Thermal impedance ($Z_{\theta JX}$) is not required in screen 13.

4.3.1 JAN testing. JAN level product will have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening level requirements.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081, as applicable, of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). (See table II, subgroup 4).

4.3.3 Power burn-in conditions. Power burn-in conditions are as follows: $I_{ZM(min)}$ = column 8 of table IV; T_A = 75°C maximum. Test conditions in accordance with method 1038 of MIL-STD-750, condition B (see 4.5.6). Adjust I_Z or T_A to achieve the required T_J. T_J = 125°C minimum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table III herein.

Subgroup	Method	Condition
B2	1056	0°C to +100°C, 10 cycles.
B2	1051	-55°C to +175°C, 25 cycles.
B2	4066	Condition B, See 4.5.1.
B3	1027	$I_{Z(min)}$ = column 8 of table IV. Adjust I_Z or T_A to achieve T_J = 150°C minimum (see 4.5.6).
B4	2101	Decap analysis, scribe and break only.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table III herein.

Subgroup	Method	Condition
C2	1056	0°C to +100°C, 10 cycles.
C2	1051	-55°C to +175°C, 25 cycles.
C2	2036	Tension: Test condition A; weight = 10 pounds (4.54 Kg), t = 15 seconds. Lead fatigue: Test condition E. (Lead fatigue is not required for UR-1, UB, UB2 and UD suffix devices)
C2	1071	Test condition E.
C5	4081	See 4.3.2 herein.
C6	1027	$I_{Z(min)}$ = column 8 of table IV. Adjust I_Z or T_A to achieve T_J = 150°C minimum (see 4.5.6).
C7		Not applicable.
C8	4071	I_Z = column 5 of table IV; T_1 = +25°C ±5°C, T_2 = +125°C ±5°C. (Maximum limit in accordance with column 14 of table IV). Sample size = 22, c = 0. (See 4.5.4.)

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with appendix E, table E-IX of MIL-PRF-19500 and table II herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table III herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Surge current (I_{ZSM}). The peak currents shown in column 10 of table IV shall be applied in the reverse direction and these shall be imposed on the current ($I_Z = I_{Z1}$) (column 5 of table IV) a total of 5 surges at 1-minute intervals. Each individual surge shall be one-half square-wave-pulse of one one-hundred twenty second duration or an equivalent one-half sine wave with the same effective rms current.

4.5.2 Regulator voltage measurements. The test current shall be applied until thermal equilibrium is attained (90 seconds minimum) prior to reading the breakdown voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located at .375 inch (9.53 mm) from the body (UR version = 0 lead length) and the mounting clips shall be maintained at a temperature of +25°C +8°C, and -2°C. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

4.5.3 Voltage regulation $V_{Z(reg)}$. Voltage regulation shall be determined by the difference of the regulator voltage measured at different currents as specified in table I, subgroup 7 herein. Both tests shall be performed at thermal equilibrium. This ΔV_Z shall not exceed column 9 of table IV.

4.5.4 Temperature coefficient of regulator voltage (αV_Z). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified ambient temperature as specified in 4.4.3, subgroup C8.

4.5.5 Scope display evaluation. Scope display evaluation shall be stable in accordance with method 4023 of MIL-STD-750, condition A. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. The reverse current over the knee shall be 500 µA peak.

4.5.6 Free air burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees as a minimum the $I_{Z(min)}$ described in 4.3.3 and that the minimum applied voltage, where applicable, is maintained through-out the burn-in period. Use method 3100 of MIL-STD-750 to measure T_J.

4.6 Element evaluation of un-encapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit 2/	Limit 2/	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 3/	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
Forward voltage	4011	Condition B, I_F = 200 mA dc.	V_F			V dc
1N962 - 1N985 4/ 1N986 - 1N992 4/					1.1 1.3	
Reverse current	4016	DC method, Condition A or B, V_R = column 11 of table IV.	I_{R1}		Col. 12	µA dc
Regulator voltage	4022	I_{Z1} = column 5 of table IV, see 4.5.2	V_Z	Col. 3	Col. 4	V dc
Subgroup 3						
High temperature operation		T_A = 150°C				
Reverse current	4016	DC method, Condition A or B, V_R = column 11 of table IV.	I_{R2}		Col. 13	µA dc
Subgroup 4						
Small-signal reverse breakdown impedance	4051	I_Z = column 5 of table IV. I_{SIG} = 10 percent of I_Z rms.	Z_Z		Col. 6	ohm
Small-signal reverse breakdown impedance	4051	I_{ZK} = 250 µA dc, I_{SIG} = 25 µA rms.	Z_{ZK}		Col. 7	ohm
Scope display	4023	See 4.5.5, n = 116, c = 0.				
Subgroups 5						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit 2/	Limit 2/	Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Surge current	4066	Condition B, See 4.5.1				
Electrical measurements		See table III, steps 1, 3, and 4.				
Subgroup 7						
Voltage regulation (see 4.5.3)		I_Z = 10 percent of column 8 of table IV (current 1). I_Z = 50 percent of column 8 of table IV (current 2).	$\Delta V_{Z(reg)}$		Col. 9	V dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table IV.
3/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JAN, JANTX, JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.
4/ Applies to all suffix versions.

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TABLE II. Group E inspection qualification and requalification (all product assurance levels).

Inspection 1/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	500 cycles	
Electrical measurements		See table III, steps 1, 2, 3, 4, and 5	
Subgroup 2			45 devices, c = 0
Intermittent life	1037	6,000 cycles. I_Z = column 8 of table IV	
Electrical measurements		See table III, steps 2, 3, 4, and 5	
Subgroup 4			
Thermal impedance curve		See MIL-PRF-19500	
Subgroups 5 and 6			
Not applicable			
Subgroup 9			n = 45
Resistance to glass cracking	1057	Condition B. Cool down after solder immersion is permitted. Test until failure occurs on all devices with the chosen sample or to a maximum of 25 cycles, whichever comes first	

1/ A separate sample may be pulled for each test.

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TABLE III. Group B, C, and E electrical and delta end-point measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limit	Limit	Unit
		Method	Conditions		Min	Max	
1.	Reverse current	4016	DC method; Condition A or B, V_R = column 11 of table IV.	I_{R1}		Column 12	µA dc
2.	Reverse current	4016	DC method; Condition A or B, V_R = column 11 of table IV.	I_{R3}		2	µA dc
3.	Regulator voltage (see 4.5.2)	4022	I_{Z1} = column 5 of table IV.	V_Z	Column 3	Column 4	V dc
4.	Small-signal breakdown impedance	4051	I_{Z1} = column 5 of table IV. I_{SIG} = 10 percent of I_Z rms	Z_{ZT}		Column 6	Ohms
5.	Thermal impedance	3101	See 4.3.2	$\Delta Z_{\theta JX}$		10 percent of initial value max.	°C/W

1/ The electrical measurements for group B inspection (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1, 3, 4, and 5 of this table shall also be taken.
   b. In addition to the measurements specified for subgroups 3 and 6, the measurements of steps 2, 3, and 4 of this table shall also be taken.

2/ The electrical measurements for group C inspection (table E-VII of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of steps 1, 3, 4, and 5 of this table shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of steps 2, 3, and 4 of this table shall also be taken.

3/ The electrical measurements for group E inspection (table E-IX of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 1, the measurements of steps 1, 2, 3, 4, and 5 of this table shall also be taken .
   b. In addition to the measurements specified for subgroup 2, the measurements of steps 2, 3, 4, and 5 of this table shall also be taken.

4/ Column references are to table IV.

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TABLE IV. Electrical characteristics (5, 2 and 1 percent tolerance diodes).

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 10	Col 11	Col 12	Col 13	Col 14
Type	V_Z nom	V_Z min	V_Z max	I_{Z1}	Z_Z	Z_{ZK}	I_{ZM}	ΔV_Z	I_{ZSM}	I_{ZSM} (surge) UB	V_R	I_{R1} at T_A = +25°C V_R = col 11	I_{R2} at T_A = +150°C V_R = col 11	α_{VZ}
	Volts	Volts	Volts	mA	Ω	Ω	mA	Volts	mA	mA	Volts	µA	µA	%/°C
1N962B	11	10.45	11.55	11.5	9.5	700	35	0.50	590	184	8.4	1.0	10	+0.073
1N963B	12	11.40	12.60	10.5	11.5	700	32	0.55	540	169	9.1	1.0	10	+0.076
1N964B	13	12.35	13.65	9.5	13.0	700	30	0.60	500	156	9.9	0.5	10	+0.079
1N965B	15	14.25	15.75	8.5	16.0	700	26	0.70	433	135	11	0.5	10	+0.082
1N966B	16	15.20	16.80	7.8	17.0	700	25	0.75	406	127	12	0.5	10	+0.083
1N967B	18	17.10	18.90	7.0	21	750	21	0.85	361	113	14	0.5	10	+0.085
1N968B	20	19.0	21.0	6.2	25	750	19	0.95	325	102	15	0.5	10	+0.086
1N969B	22	20.9	23.1	5.6	29	750	17	1.05	295	93	17	0.5	10	+0.087
1N970B	24	22.8	25.2	5.2	33	750	16	1.15	271	85	18	0.5	10	+0.088
1N971B	27	25.7	28.3	4.6	41	750	14	1.30	240	75	21	0.5	10	+0.090
1N972B	30	28.5	31.5	4.2	49	1000	13	1.45	216	68	23	0.5	10	+0.091
1N973B	33	31.4	34.6	3.8	58	1000	12	1.60	197	62	25	0.5	10	+0.092
1N974B	36	34.2	37.8	3.4	70	1000	11	1.75	180	56	27	0.5	10	+0.093
1N975B	39	37.1	40.9	3.2	80	1000	9.1	1.90	166	52	30	0.5	10	+0.094
1N976B	43	40.9	45.1	3.0	93	1000	8.8	2.10	151	47	33	0.5	10	+0.095
1N977B	47	44.7	49.3	2.7	105	1500	7.9	2.25	138	43	36	0.5	10	+0.095
1N978B	51	48.5	53.5	2.5	125	1500	7.4	2.5	127	40	39	0.5	10	+0.096
1N979B	56	53.2	58.8	2.2	150	2000	6.9	2.7	116	36	43	0.5	10	+0.096
1N980B	62	58.9	65.1	2.0	185	2000	6.0	2.9	105	33	47	0.5	10	+0.097
1N981B	68	64.6	71.4	1.8	230	2000	5.5	3.2	95	30	52	0.5	10	+0.097
1N982B	75	71.3	78.7	1.7	270	2000	5.1	3.4	86	27	56	0.5	10	+0.098
1N983B	82	77.9	86.1	1.5	330	3000	4.6	3.8	79	35	62	0.5	10	+0.098
1N984B	91	86.5	95.5	1.4	400	3000	4.2	4.2	71	22	69	0.5	10	+0.099
1N985B	100	95.0	105.0	1.3	500	3000	3.7	4.4	65	20	76	0.5	10	+0.110
1N986B	110	104.5	115.5	1.1	750	4000	3.3	4.8	59	18	84	0.5	10	+0.110
1N987B	120	114.0	126.0	1.0	900	4500	3.1	5.2	54	17	91	0.5	10	+0.110
1N988B	130	123.5	136.5	0.95	1100	5000	2.7	5.6	50	16	99	0.5	10	+0.110
1N989B	150	142.5	157.5	0.85	1500	6000	2.4	7.0	43	13	114	0.5	10	+0.110
1N990B	160	152.0	168.0	0.80	1700	6500	2.2	7.5	40	12.5	122	0.5	10	+0.110
1N991B	180	171.0	189.0	0.68	2200	7100	2.0	9.0	36	11	137	0.5	10	+0.110
1N992B	200	190.0	210.0	0.65	2500	8000	1.8	12.0	32	10	152	0.5	10	+0.110

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TABLE IV. Electrical characteristics (5, 2 and 1 percent tolerance diodes).

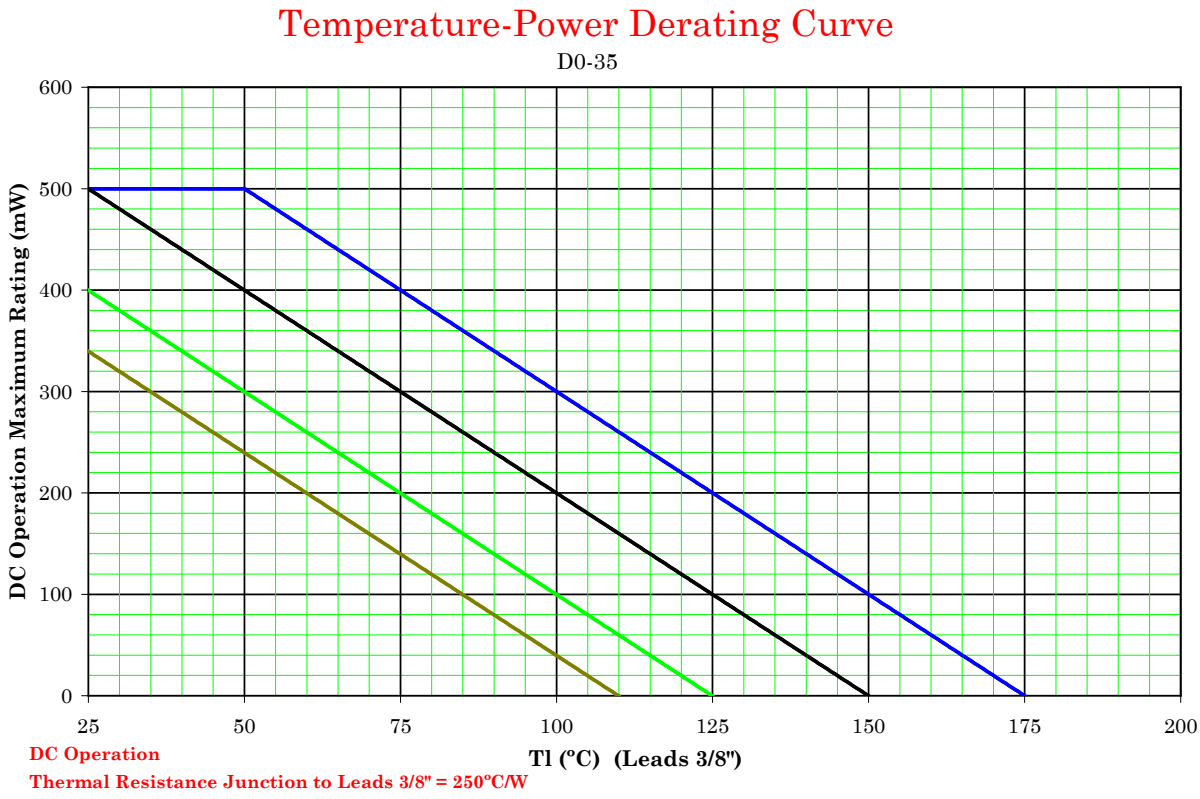
Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 10	Col 11	Col 12	Col 13	Col 14
Type	V_Z nom	V_Z min	V_Z max	I_{Z1}	Z_Z	Z_{ZK}	I_{ZM}	ΔV_Z	I_{ZSM}	I_{ZSM} (surge) UB	V_R	I_{R1} at T_A = +25°C V_R = col 11	I_{R2} at T_A = +150°C V_R = col 11	α_{VZ}
	Volts	Volts	Volts	mA	Ω	Ω	mA	Volts	mA	mA	Volts	µA	µA	%/°C
1N962C	11	10.78	11.22	11.5	9.5	700	35	0.50	590	184	8.4	1.0	10	+0.073
1N963C	12	11.76	12.24	10.5	11.5	700	32	0.55	540	169	9.1	1.0	10	+0.076
1N964C	13	12.74	13.25	9.5	13.0	700	30	0.60	500	156	9.9	0.5	10	+0.079
1N965C	15	14.70	15.30	8.5	16.0	700	26	0.70	433	135	11	0.5	10	+0.082
1N966C	16	15.68	16.32	7.8	17.0	700	25	0.75	406	127	12	0.5	10	+0.083
1N967C	18	17.64	18.36	7.0	21	750	21	0.85	361	113	14	0.5	10	+0.085
1N968C	20	19.60	20.40	6.2	25	750	19	0.95	325	102	15	0.5	10	+0.086
1N969C	22	21.56	22.44	5.6	29	750	17	1.05	295	93	17	0.5	10	+0.087
1N970C	24	23.52	24.48	5.2	33	750	16	1.15	271	85	18	0.5	10	+0.088
1N971C	27	26.46	27.54	4.6	41	750	14	1.30	240	75	21	0.5	10	+0.090
1N972C	30	29.40	30.60	4.2	49	1000	13	1.45	216	68	23	0.5	10	+0.091
1N973C	33	32.34	33.66	3.8	58	1000	12	1.60	197	62	25	0.5	10	+0.092
1N974C	36	35.28	36.72	3.4	70	1000	11	1.75	180	56	27	0.5	10	+0.093
1N975C	39	38.22	39.78	3.2	80	1000	9.1	1.90	166	52	30	0.5	10	+0.094
1N976C	43	42.14	43.86	3.0	93	1000	8.8	2.10	151	47	33	0.5	10	+0.095
1N977C	47	46.06	47.94	2.7	105	1500	7.9	2.25	138	43	36	0.5	10	+0.095
1N978C	51	49.98	52.02	2.5	125	1500	7.4	2.5	127	40	39	0.5	10	+0.096
1N979C	56	54.88	57.12	2.2	150	2000	6.9	2.7	116	36	43	0.5	10	+0.096
1N980C	62	60.76	63.24	2.0	185	2000	6.0	2.9	105	33	47	0.5	10	+0.097
1N981C	68	66.64	69.36	1.8	230	2000	5.5	3.2	95	30	52	0.5	10	+0.097
1N982C	75	73.50	76.50	1.7	270	2000	5.1	3.4	86	27	56	0.5	10	+0.098
1N983C	82	80.36	83.64	1.5	330	3000	4.6	3.8	79	35	62	0.5	10	+0.098
1N984C	91	89.18	92.82	1.4	400	3000	4.2	4.2	71	22	69	0.5	10	+0.099
1N985C	100	98.0	102.0	1.3	500	3000	3.7	4.4	65	20	76	0.5	10	+0.110
1N986C	110	107.8	112.2	1.1	750	4000	3.3	4.8	59	18	84	0.5	10	+0.110
1N987C	120	117.6	122.4	1.0	900	4500	3.1	5.2	54	17	91	0.5	10	+0.110
1N988C	130	127.4	132.6	0.95	1100	5000	2.7	5.6	50	16	99	0.5	10	+0.110
1N989C	150	147.0	153.0	0.85	1500	6000	2.4	7.0	43	13	114	0.5	10	+0.110
1N990C	160	156.8	163.2	0.80	1700	6500	2.2	7.5	40	12.5	122	0.5	10	+0.110
1N991C	180	176.4	183.6	0.68	2200	7100	2.0	9.0	36	11	137	0.5	10	+0.110
1N992C	200	196.0	204.0	0.65	2500	8000	1.8	12.0	32	10	152	0.5	10	+0.110

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TABLE IV. Electrical characteristics (5, 2, and 1 percent tolerance diodes).

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 10	Col 11	Col 12	Col 13	Col 14
Type	V_Z nom	V_Z min	V_Z max	I_{Z1}	Z_Z	Z_{ZK}	I_{ZM}	ΔV_Z	I_{ZSM}	I_{ZSM} (surge) UB	V_R	I_{R1} at T_A = +25°C V_R = col 11	I_{R2} at T_A = +150°C V_R = col 11	α_{VZ}
	Volts	Volts	Volts	mA	Ω	Ω	mA	Volts	mA	mA	Volts	µA	µA	%/°C
1N962D	11	10.89	11.11	11.5	9.5	700	35	0.50	590	184	8.4	1.0	10	+0.073
1N963D	12	11.88	12.12	10.5	11.5	700	32	0.55	540	169	9.1	1.0	10	+0.076
1N964D	13	12.87	13.13	9.5	13.0	700	30	0.60	500	156	9.9	0.5	10	+0.079
1N965D	15	14.85	15.15	8.5	16.0	700	26	0.70	433	135	11	0.5	10	+0.082
1N966D	16	15.84	16.16	7.8	17.0	700	25	0.75	406	127	12	0.5	10	+0.083
1N967D	18	17.82	18.18	7.0	21	750	21	0.85	361	113	14	0.5	10	+0.085
1N968D	20	19.80	20.20	6.2	25	750	19	0.95	325	102	15	0.5	10	+0.086
1N969D	22	21.78	22.22	5.6	29	750	17	1.05	295	93	17	0.5	10	+0.087
1N970D	24	23.76	24.24	5.2	33	750	16	1.15	271	85	18	0.5	10	+0.088
1N971D	27	26.73	27.27	4.6	41	750	14	1.30	240	75	21	0.5	10	+0.090
1N972D	30	29.70	30.30	4.2	49	1000	13	1.45	216	68	23	0.5	10	+0.091
1N973D	33	32.67	33.33	3.8	58	1000	12	1.60	197	62	25	0.5	10	+0.092
1N974D	36	35.64	36.36	3.4	70	1000	11	1.75	180	56	27	0.5	10	+0.093
1N975D	39	38.61	39.39	3.2	80	1000	9.1	1.90	166	52	30	0.5	10	+0.094
1N976D	43	42.57	43.43	3.0	93	1000	8.8	2.10	151	47	33	0.5	10	+0.095
1N977D	47	46.53	47.47	2.7	105	1500	7.9	2.25	138	43	36	0.5	10	+0.095
1N978D	51	50.49	51.51	2.5	125	1500	7.4	2.5	127	40	39	0.5	10	+0.096
1N979D	56	55.44	56.56	2.2	150	2000	6.9	2.7	116	36	43	0.5	10	+0.096
1N980D	62	61.38	62.62	2.0	185	2000	6.0	2.9	105	33	47	0.5	10	+0.097
1N981D	68	67.32	68.68	1.8	230	2000	5.5	3.2	95	30	52	0.5	10	+0.097
1N982D	75	74.25	75.75	1.7	270	2000	5.1	3.4	86	27	56	0.5	10	+0.098
1N983D	82	81.18	82.82	1.5	330	3000	4.6	3.8	79	35	62	0.5	10	+0.098
1N984D	91	90.09	91.91	1.4	400	3000	4.2	4.2	71	22	69	0.5	10	+0.099
1N985D	100	99.0	101.0	1.3	500	3000	3.7	4.4	65	20	76	0.5	10	+0.110
1N986D	110	108.9	111.1	1.1	750	4000	3.3	4.8	59	18	84	0.5	10	+0.110
1N987D	120	118.8	121.2	1.0	900	4500	3.1	5.2	54	17	91	0.5	10	+0.110
1N988D	130	128.7	131.3	0.95	1100	5000	2.7	5.6	50	16	99	0.5	10	+0.110
1N989D	150	148.5	151.5	0.85	1500	6000	2.4	7.0	43	13	114	0.5	10	+0.110
1N990D	160	158.4	161.6	0.80	1700	6500	2.2	7.5	40	12.5	122	0.5	10	+0.110
1N991D	180	178.2	181.8	0.68	2200	7100	2.0	9.0	36	11	137	0.5	10	+0.110
1N992D	200	198.0	202.0	0.65	2500	8000	1.8	12.0	32	10	152	0.5	10	+0.110

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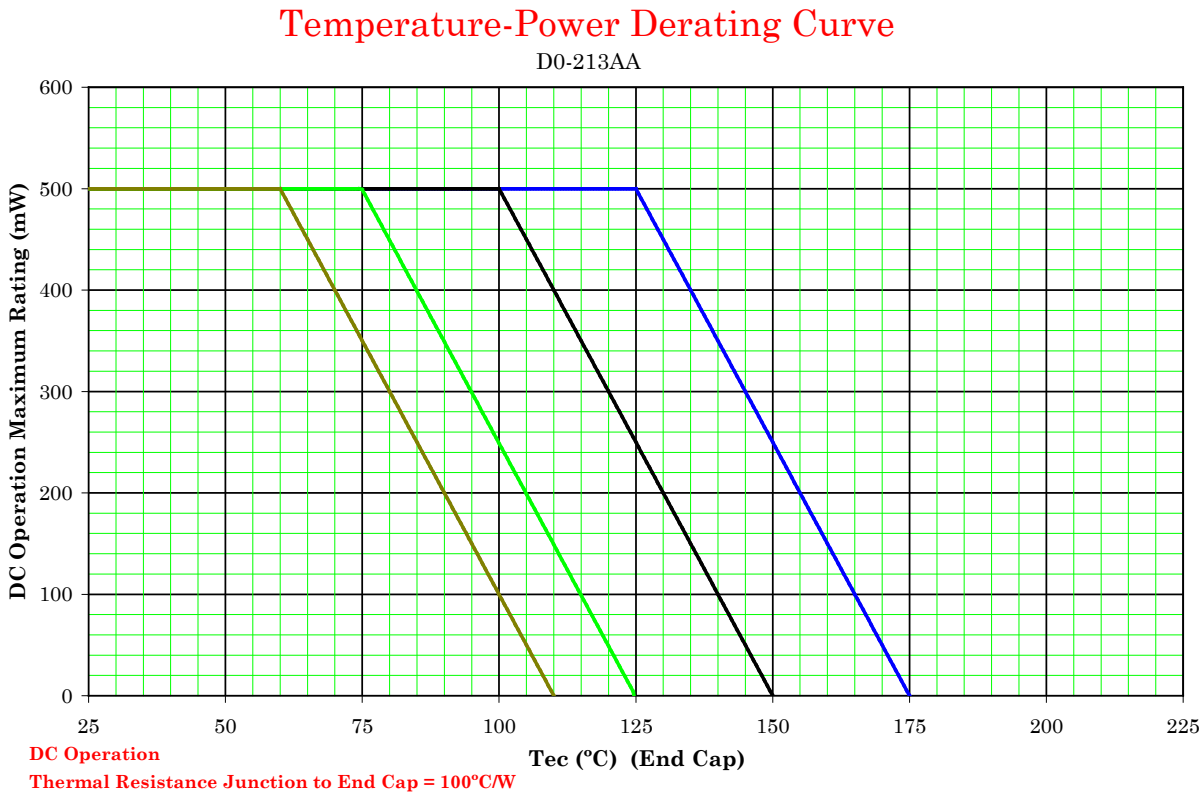


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 175°C) and power rating specified (see 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 9. Temperature-power derating curve, junction to lead (T_L) for axial leaded DO-35.

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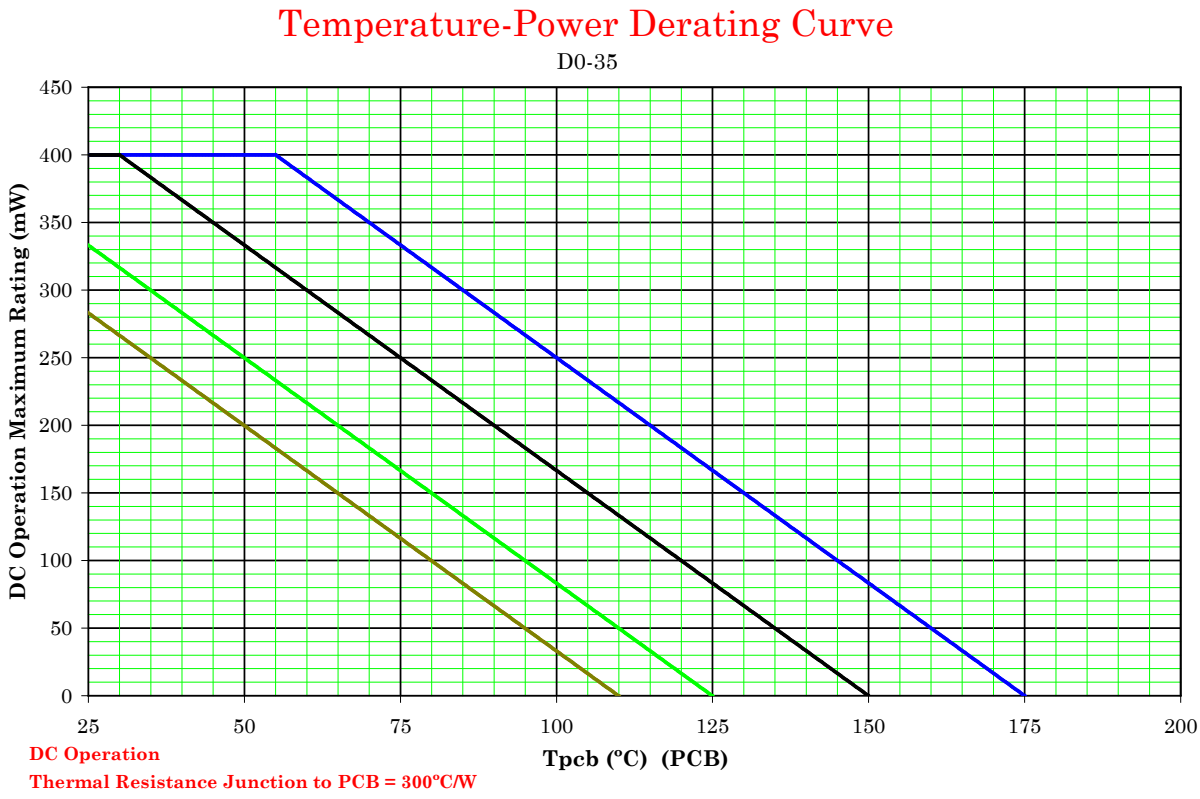


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 175°C) and power rating specified (see 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 10. Temperature-power derating curve, junction to end-cap (T_{EC}) for UR surface mount.

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NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 175°C) and power rating specified (see 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 11. Temperature-power derating curve, junction to PCB (T_{PCB}) for axial leaded DO-35.

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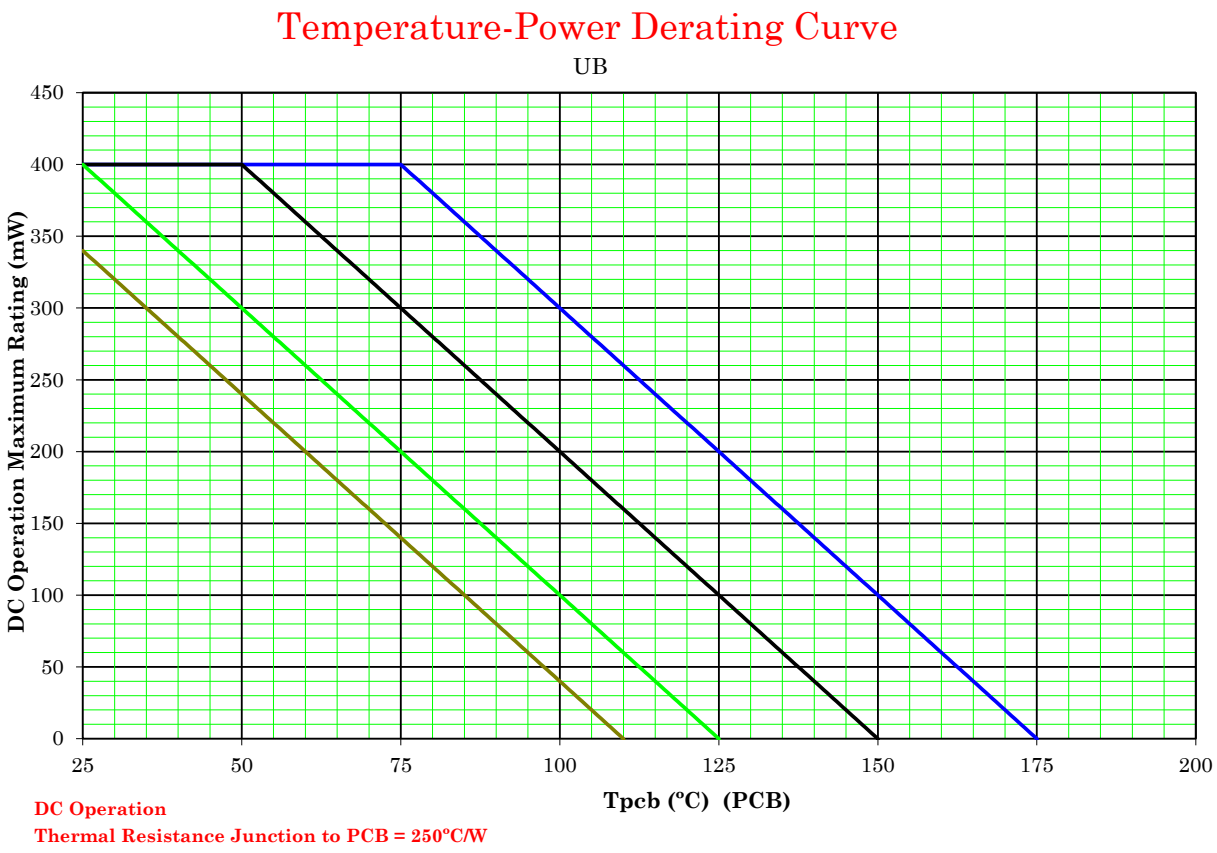


FIGURE 12. Temperature-power derating curve, junction to PCB (T_{PCB}) for UB package.

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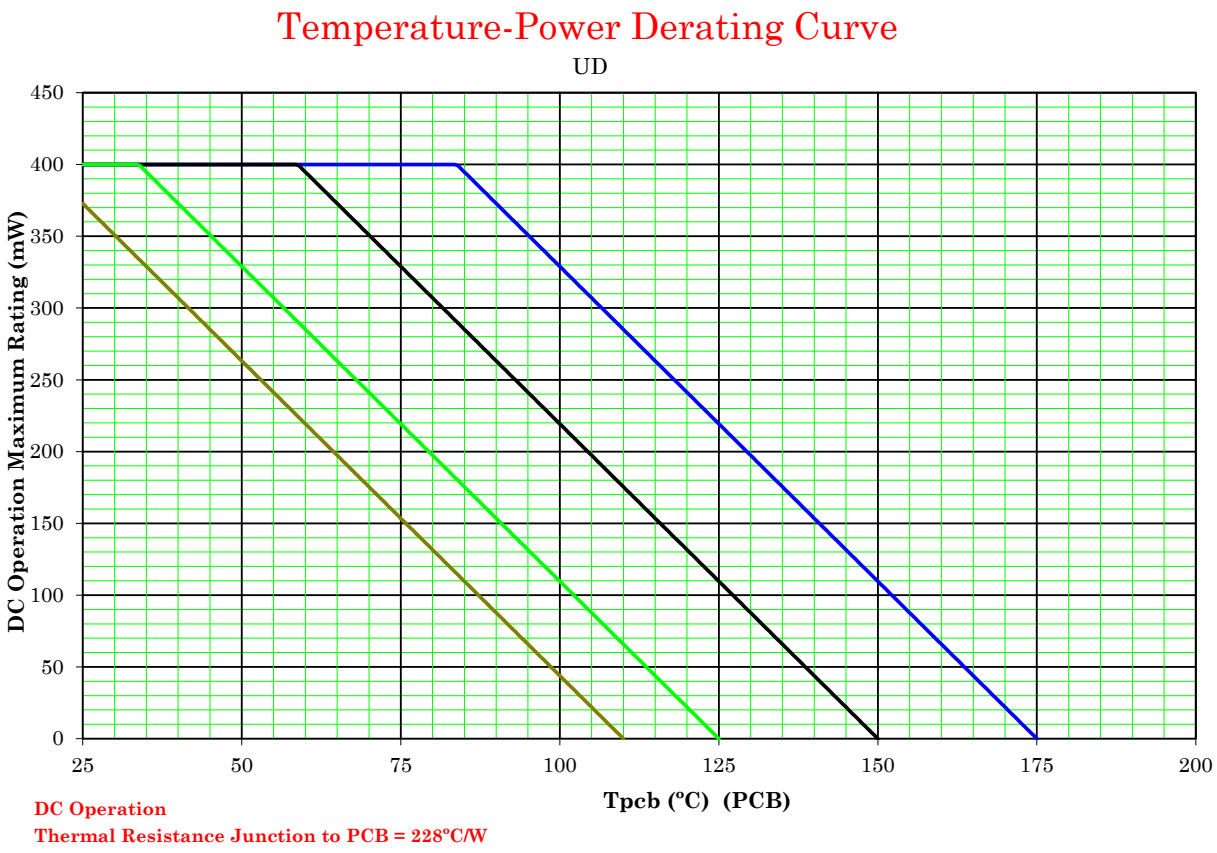
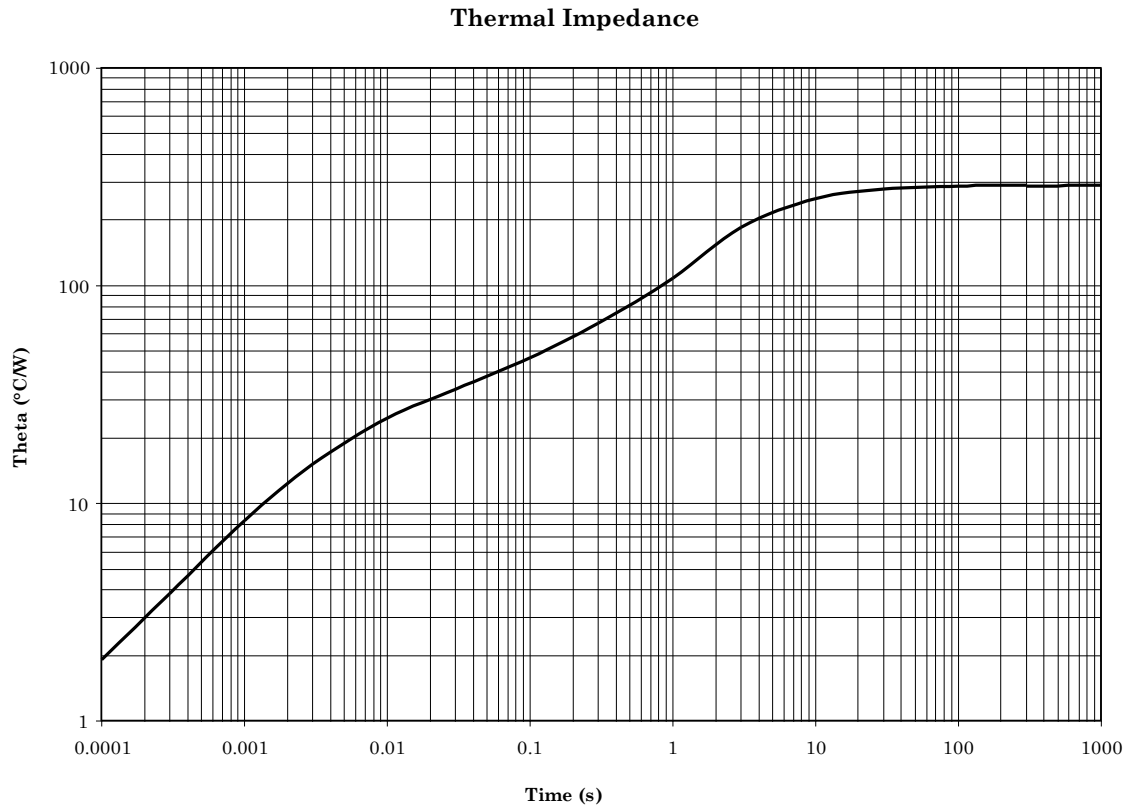


FIGURE 13. Temperature-power derating curve, junction to PCB (T_{PCB}) for UD package.

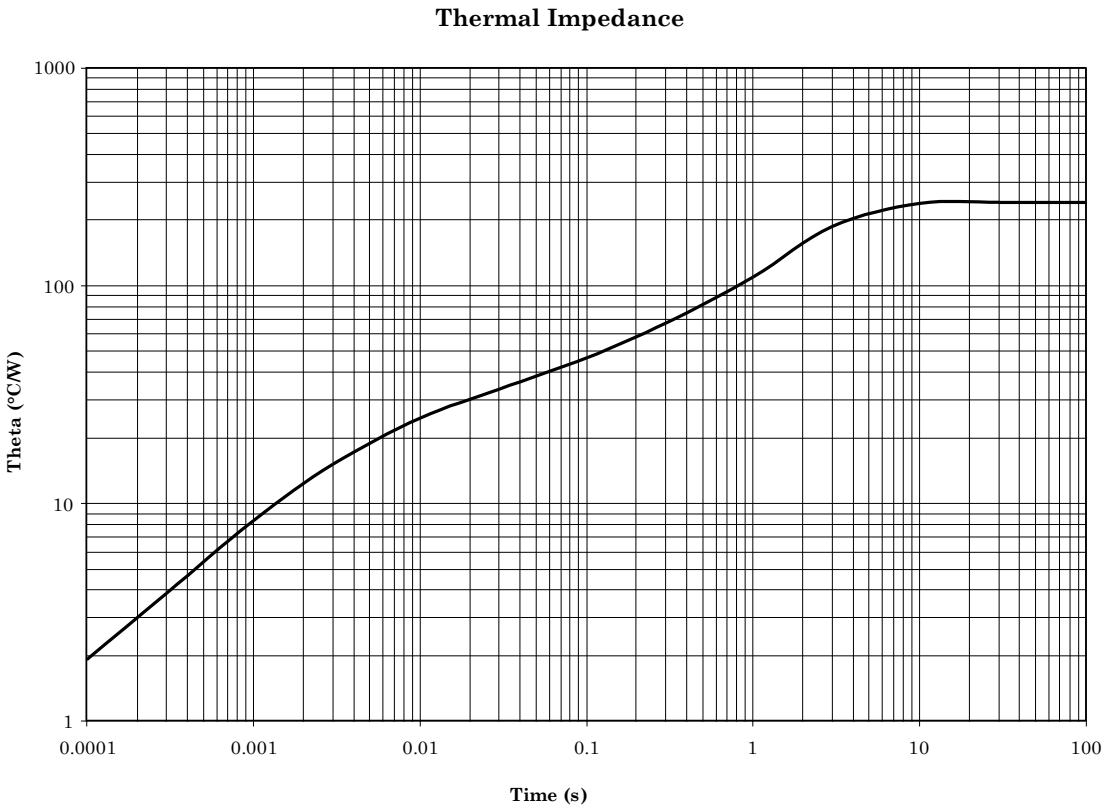
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:15Z
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NOTES:
1. Thermal impedance DO-35 PCB mount, FR4, 1oz Cu, .050x .087 inch (1.27 x 2.21 mm) pad (MELF) and .092 inch (2.33 mm) diameter (axial with .125 inch (3.175 mm) lead length) at T_A = 25°C.
2. Thermal resistance = 300 °C/W. Maximum power rating = 400 mW at T_A = 55°C.

FIGURE 14. Thermal impedance graph for DO-35 package (PCB mounted).

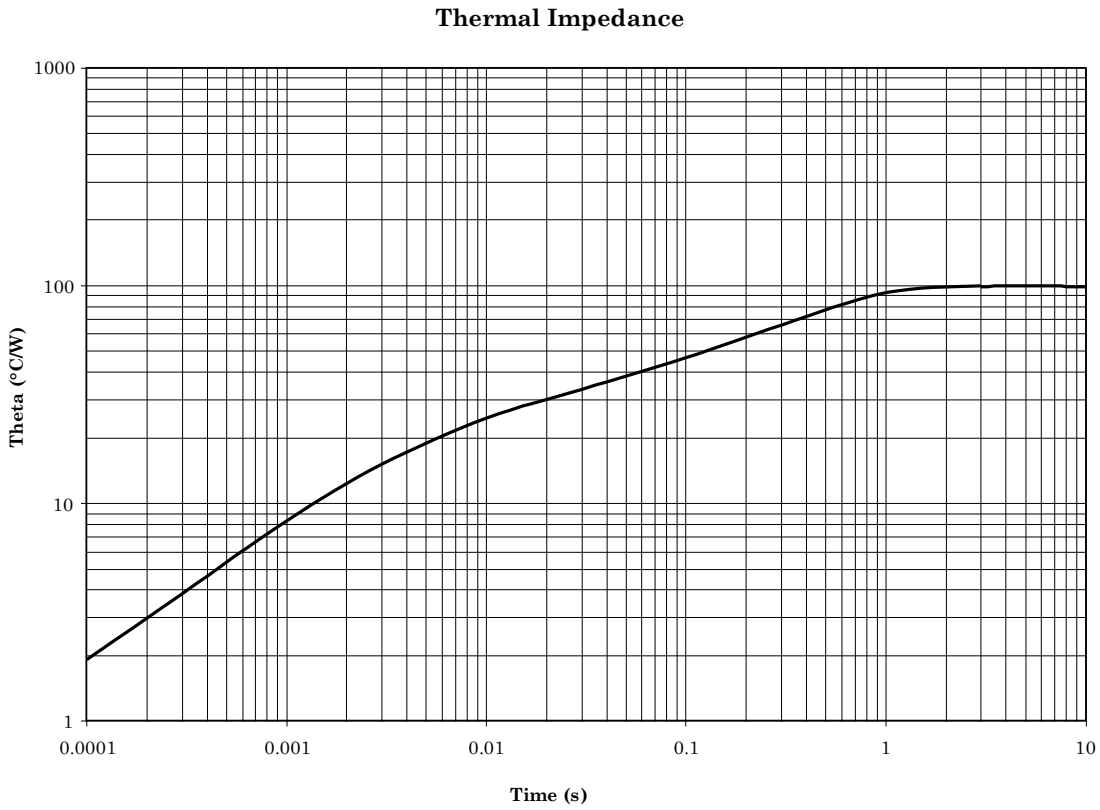
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NOTES:
1. Thermal impedance DO-35 axial, T_J = 25°C at .375 inch (9.52 mm) from body.
2. Thermal resistance = 250 °C/W. Maximum power rating = 500 mW at T_L = 50°C.

FIGURE 15. Thermal impedance graph for DO-35 axial package.

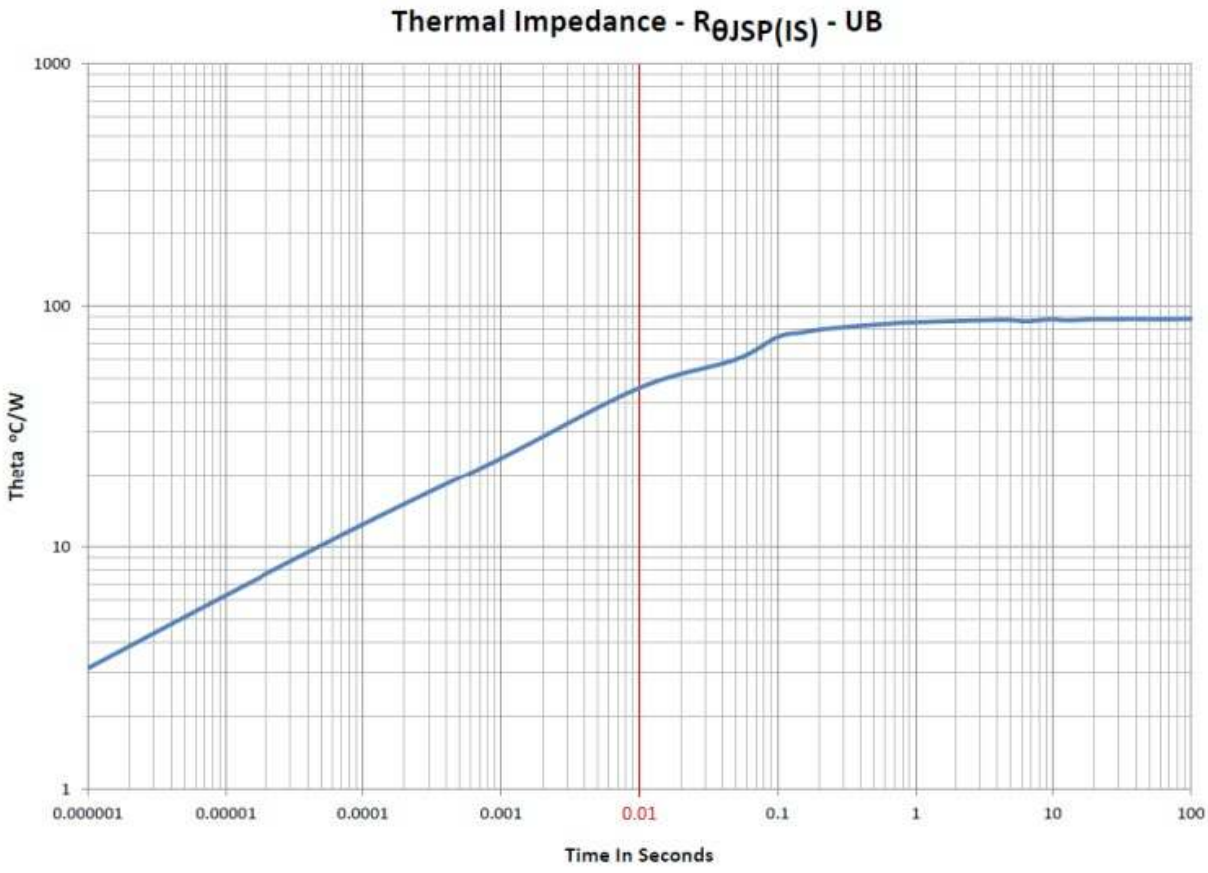
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NOTES:
1. Thermal impedance DO-213AA MELF, T_{EC} =25°C.
2. Thermal resistance = 100°C/W. Power rating = 500 mW at T_{EC} = 125°C.

FIGURE 16. Thermal impedance graph for DO-213AA MELF package.

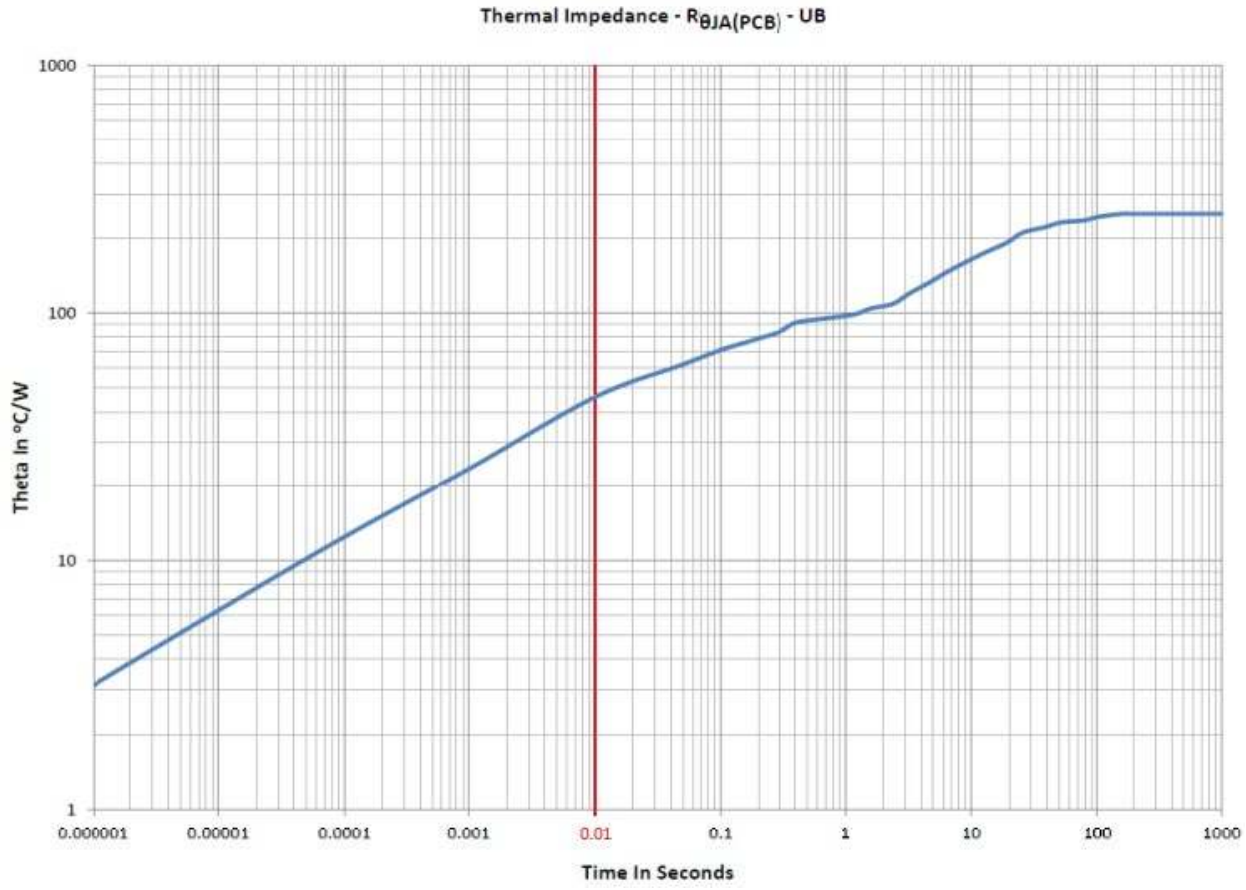
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NOTES:
1. Thermal impedance $Z_{\theta JX}$ = 45°C/W maximum at 10 ms.
2. Thermal resistance $R_{\theta JSP(IS)}$ = 90°C/W.

FIGURE 17 Thermal impedance graph $R_{\theta JSP(IS)}$ for UB package.

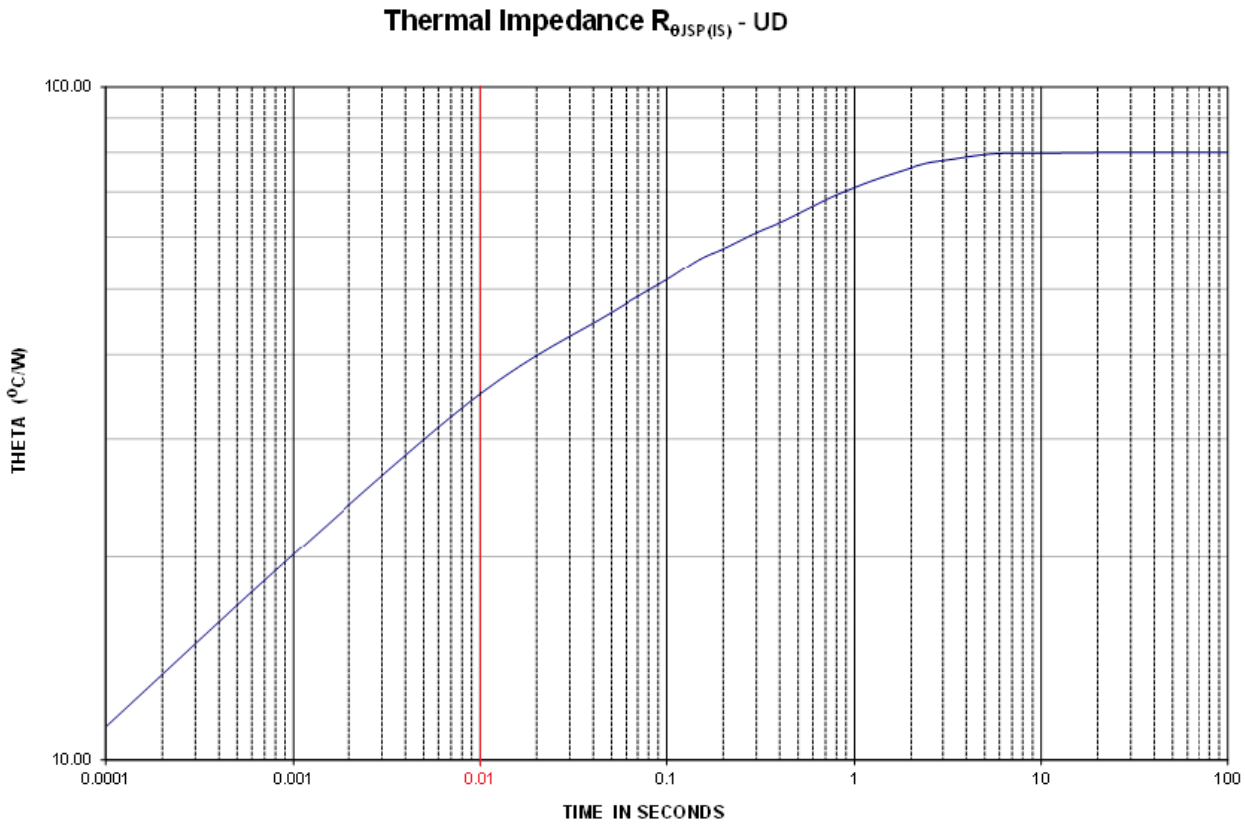
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:15Z
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NOTES:
1. Thermal impedance $Z_{\theta JX}$ = 45°C/W maximum at 10 ms.
2. Thermal resistance $R_{\theta JA(PCB)}$ = 250°C/W.

FIGURE 18. Thermal impedance graph $R_{\theta JA(PCB)}$ for UB package.

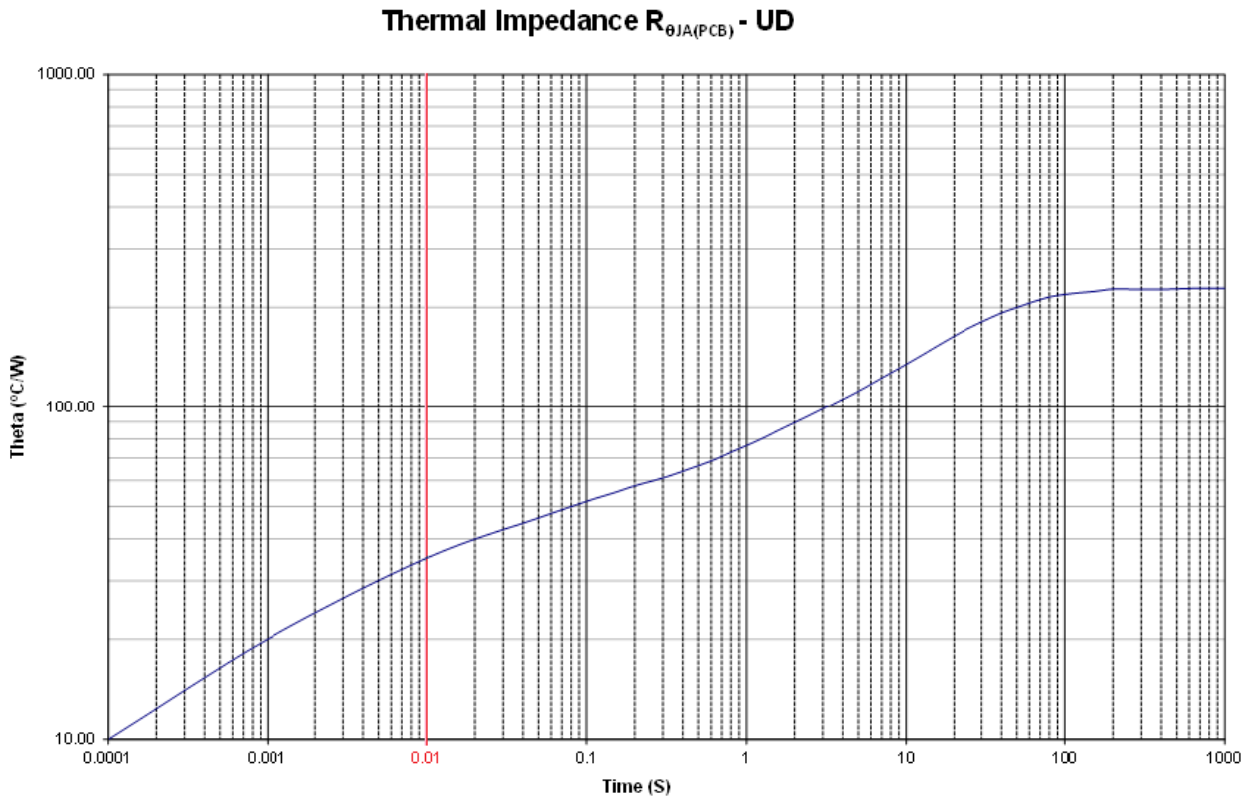
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NOTES:
1. Thermal impedance $Z_{\theta JX}$ = 35°C/W maximum at 10 ms.
2. Thermal resistance $R_{\theta JSP(IS)}$ = 80°C/W.

FIGURE 19. Thermal impedance graph $R_{\theta JSP(IS)}$ for UD package.

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NOTES:
1. Thermal impedance $Z_{\theta JX}$ = 35°C/W maximum at 10 ms.
2. Thermal resistance $R_{\theta JA(PCB)}$ = 228°C/W.

FIGURE 20. Thermal impedance graph $R_{\theta JA(PCB)}$ for UD package.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

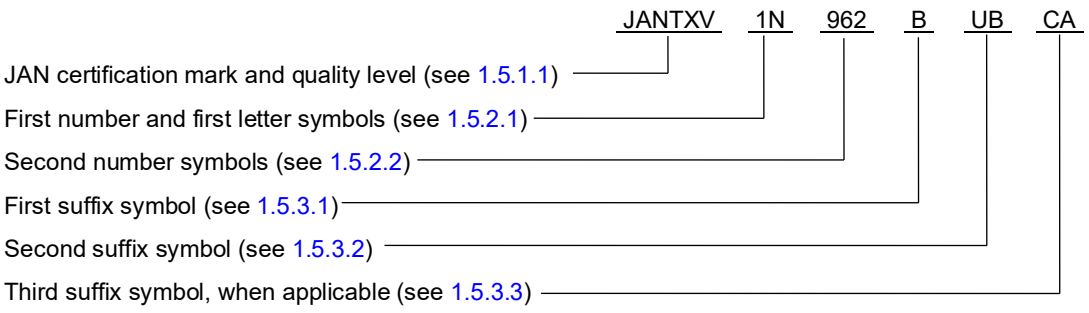
d. Product assurance level and type designator.

e. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Weapons Support, VQE, P.O. Box 3990, Columbus, OH 43216-5000 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

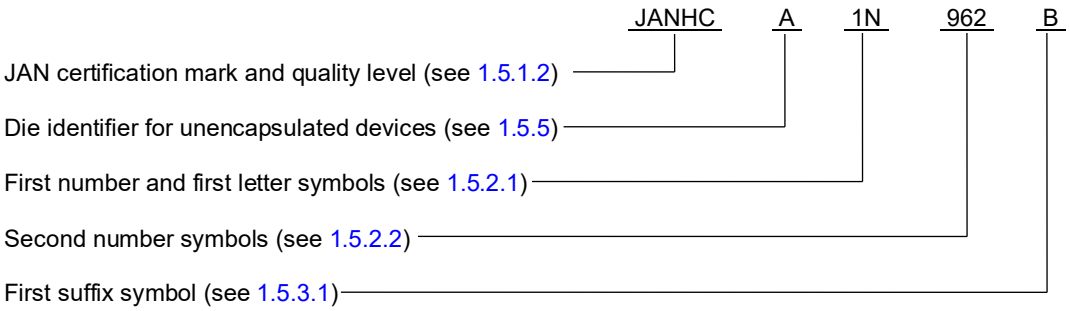
6.4 PIN construction examples.

6.4.1 Encapsulated devices. The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

6.5.1 List of encapsulated device types.

PINs for type 1N962 through 1N992 1/	
JAN1N962B-1 through JAN1N992B-1	JANTX1N962BUB2# through JANTX1N992BUB2#
JAN1N962C-1 through JAN 1N992C-1	JANTX1N962CUB2# through JANTX1N992CUB2#
JAN1N962D-1 through JAN1N992D-1	JANTX1N962DUB2# through JANTX1N992DUB2#
JAN1N962BUR-1 through JAN1N992BUR-1	JANTX1N962BUD through JANTX1N992BUD
JAN1N962CUR-1 through JAN1N992CUR-1	JANTX1N962CUD through JANTX1N992CUD
JAN1N962DUR-1 through JAN1N992DUR-1	JANTX1N962DUD through JANTX1N992DUD
JAN1N962BUB@ through JAN1N992BUB@	JANTXV1N962BUR-1 through JANTXV1N992BUR-1
JAN1N962CUB@ through JAN1N992CUB@	JANTXV1N962CUR-1 through JANTXV1N992CUR-1
JAN1N962DUB@ through JAN1N992DUB@	JANTXV1N962DUR-1 through JANTXV1N992DUR-1
JAN1N962BUB2# through JAN1N992BUB2#	JANTXV1N962B-1 through JANTXV1N992B-1
JAN1N962CUB2# through JAN1N992CUB2#	JANTXV1N962C-1 through JANTXV1N992C-1
JAN1N962DUB2# through JAN1N992DUB2#	JANTXV1N962D-1 through JANTXV1N992D-1
JAN1N962BUD through JAN1N992BUD	JANTXV1N962BUB@ through JANTXV1N992BUB@
JAN1N962CUD through JAN1N992CUD	JANTXV1N962CUB@ through JANTXV1N992CUB@
JAN1N962DUD through JAN1N992DUD	JANTXV1N962DUB@ through JANTXV1N992DUB@
JANTX1N962B-1 through JANTX1N992B-1	JANTXV1N962BUB2# through JANTXV1N992BUB2#
JANTX1N962C-1 through JANTX1N992C-1	JANTXV1N962CUB2# through JANTXV1N992CUB2#
JANTX1N962D-1 through JANTX1N992D-1	JANTXV1N962DUB2# through JANTXV1N992DUB2#
JANTX1N962BUR-1 through JANTX1N992BUR-1	JANTXV1N962BUD through JANTXV1N992BUD
JANTX1N962CUR-1 through JANTX1N992CUR-1	JANTXV1N962CUD through JANTXV1N992CUD
JANTX1N962DUR-1 through JANTX1N992DUR-1	JANTXV1N962DUD through JANTXV1N992DUD
JANTX1N962BUB@ through JANTX1N992BUB@	
JANTX1N962CUB@ through JANTX1N992CUB@	
JANTX1N962DUB@ through JANTX1N992DUB@	

1/ PINs are shown using the "B" first suffix symbol. The PIN is also available with a "C" or "D" first suffix symbol, indicating the voltage tolerance. PINs for package styles UB, UB2 and UD, the "@" and "#" are the internal schematic designator placeholders (see 1.5.3.3).

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6.5.2 List of unencapsulated device types.

JANHCA1N962B	JANHCB1N962B	JANHCC1N962B	JANHCD1N962B	JANHCE1N962B
JANHCA1N963B	JANHCB1N963B	JANHCC1N963B	JANHCD1N963B	JANHCE1N963B
JANHCA1N964B	JANHCB1N964B	JANHCC1N964B	JANHCD1N964B	JANHCE1N964B
JANHCA1N965B	JANHCB1N965B	JANHCC1N965B	JANHCD1N965B	JANHCE1N965B
JANHCA1N966B	JANHCB1N966B	JANHCC1N966B	JANHCD1N966B	JANHCE1N966B
JANHCA1N967B	JANHCB1N967B	JANHCC1N967B	JANHCD1N967B	JANHCE1N967B
JANHCA1N968B	JANHCB1N968B	JANHCC1N968B	JANHCD1N968B	JANHCE1N968B
JANHCA1N969B	JANHCB1N969B	JANHCC1N969B	JANHCD1N969B	JANHCE1N969B
JANHCA1N970B	JANHCB1N970B	JANHCC1N970B	JANHCD1N970B	JANHCE1N970B
JANHCA1N971B	JANHCB1N971B	JANHCC1N971B	JANHCD1N971B	JANHCE1N971B
JANHCA1N972B	JANHCB1N972B	JANHCC1N972B	JANHCD1N972B	JANHCE1N972B
JANHCA1N973B	JANHCB1N973B	JANHCC1N973B	JANHCD1N973B	JANHCE1N973B
JANHCA1N974B	JANHCB1N974B	JANHCC1N974B	JANHCD1N974B	JANHCE1N974B
JANHCA1N975B	JANHCB1N975B	JANHCC1N975B	JANHCD1N975B	JANHCE1N975B
JANHCA1N976B	JANHCB1N976B	JANHCC1N976B	JANHCD1N976B	JANHCE1N976B
JANHCA1N977B	JANHCB1N977B	JANHCC1N977B	JANHCD1N977B	JANHCE1N977B
JANHCA1N978B	JANHCB1N978B	JANHCC1N978B	JANHCD1N978B	JANHCE1N978B
JANHCA1N979B	JANHCB1N979B	JANHCC1N979B	JANHCD1N979B	JANHCE1N979B
JANHCA1N980B	JANHCB1N980B	JANHCC1N980B	JANHCD1N980B	JANHCE1N980B
JANHCA1N981B	JANHCB1N981B	JANHCC1N981B	JANHCD1N981B	JANHCE1N981B
JANHCA1N982B	JANHCB1N982B	JANHCC1N982B	JANHCD1N982B	JANHCE1N982B
JANHCA1N983B	JANHCB1N983B	JANHCC1N983B	JANHCD1N983B	JANHCE1N983B
JANHCA1N984B	JANHCB1N984B	JANHCC1N984B	JANHCD1N984B	JANHCE1N984B
JANHCA1N985B	JANHCB1N985B	JANHCC1N985B	JANHCD1N985B	JANHCE1N985B
JANHCA1N986B	JANHCB1N986B	JANHCC1N986B	JANHCD1N986B	JANHCE1N986B
JANHCA1N987B	JANHCB1N987B	JANHCC1N987B	JANHCD1N987B	JANHCE1N987B
JANHCA1N988B	JANHCB1N988B	JANHCC1N988B	JANHCD1N988B	JANHCE1N988B
JANHCA1N989B	JANHCB1N989B	JANHCC1N989B	JANHCD1N989B	JANHCE1N989B
JANHCA1N990B	JANHCB1N990B	JANHCC1N990B	JANHCD1N990B	JANHCE1N990B
JANHCA1N991B	JANHCB1N991B	JANHCC1N991B	JANHCD1N991B	JANHCE1N991B
JANHCA1N992B	JANHCB1N992B	JANHCC1N992B	JANHCD1N992B	JANHCE1N992B

(1) For JANKC level, replace JANHC with JANKC.
(2) PINs are shown using the "B" first suffix symbol. The PIN is also available with a "C" or "D" first suffix symbol, indicating the voltage tolerance.

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6.6 Supersession information.

6.6.1 Cross reference substitution list. JANS level will no longer be built to MIL-PRF-19500/117. Devices in stock are acceptable provided the date code does not exceed the date of 8 September 1997. Devices required for space flight applications will meet the requirements of MIL-PRF-19500/533. Existing supplies of parts can be used until existing supplies are exhausted. A PIN for PIN replacement table follows, and these devices are directly interchangeable:

JANS superseded PIN	JANS superseding PIN
1N962B-1	1N6325
1N963B-1	1N6326
1N964B-1	1N6327
1N965B-1	1N6328
1N966B-1	1N6329
1N967B-1	1N6330
1N968B-1	1N6331
1N969B-1	1N6332
1N970B-1	1N6333
1N971B-1	1N6334
1N972B-1	1N6335
1N973B-1	1N6336
1N974B-1	1N6337
1N975B-1	1N6338
1N976B-1	1N6339
1N977B-1	1N6340

JANS superseded PIN	JANS superseding PIN
1N978B-1	1N6341
1N979B-1	1N6342
1N980B-1	1N6343
1N981B-1	1N6344
1N982B-1	1N6345
1N983B-1	1N6346
1N984B-1	1N6347
1N985B-1	1N6348
1N986B-1	1N6349
1N987B-1	1N6350
1N988B-1	1N6351
1N989B-1	1N6352
1N990B-1	1N6353
1N991B-1	1N6354
1N992B-1	1N6355

6.6.2 Substitutability of dash one parts. Non-dash-one devices have been deleted from this specification. Dash-one devices are a direct substitute for non dash-one devices and are preferred. The following table shows the direct substitutability. This is for all quality levels and tolerances.

Superseded PIN	Superseding PIN
1N962B	1N962B-1
1N963B	1N963B-1
1N964B	1N964B-1
1N965B	1N965B-1
1N966B	1N966B-1
1N967B	1N967B-1
1N968B	1N968B-1
1N969B	1N969B-1
1N970B	1N970B-1
1N971B	1N971B-1
1N972B	1N972B-1
1N973B	1N973B-1
1N974B	1N974B-1
1N975B	1N975B-1
1N976B	1N976B-1
1N977B	1N977B-1

Superseded PIN	Superseding PIN
1N978B	1N978B-1
1N979B	1N979B-1
1N980B	1N980B-1
1N981B	1N981B-1
1N982B	1N982B-1
1N983B	1N983B-1
1N984B	1N984B-1
1N985B	1N985B-1
1N986B	1N986B-1
1N987B	1N987B-1
1N988B	1N988B-1
1N989B	1N989B-1
1N990B	1N990B-1
1N991B	1N991B-1
1N992B	1N992B-1

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6.6.3 Substitutability of 2 percent and 1 percent tolerance devices. Devices of tighter tolerance are a direct one-way substitute for the looser tolerance devices (example: JANTX1N964D-1 substitutes for JANTX1N964B-1).

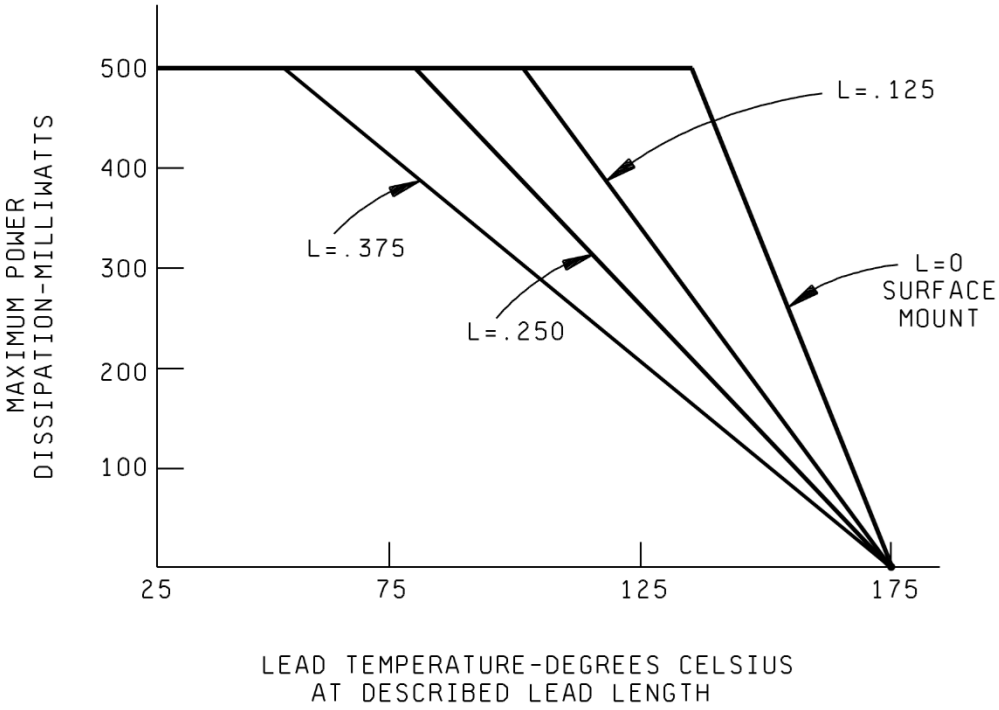
6.7 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC die suppliers with the applicable letter version (example JANHCE1N962B) will be identified on the QML.

JANHC ordering information (1)					
PIN	Manufacture CAGE				
	43611	12954	52GC4	13409	52GC4
1N962B	JANHCA1N962B	JANHCB1N962B	JANHCC1N962B	JANHCD1N962B	JANHCE1N962B
1N963B	JANHCA1N963B	JANHCB1N963B	JANHCC1N963B	JANHCD1N963B	JANHCE1N963B
1N964B	JANHCA1N964B	JANHCB1N964B	JANHCC1N964B	JANHCD1N964B	JANHCE1N964B
1N965B	JANHCA1N965B	JANHCB1N965B	JANHCC1N965B	JANHCD1N965B	JANHCE1N965B
1N966B	JANHCA1N966B	JANHCB1N966B	JANHCC1N966B	JANHCD1N966B	JANHCE1N966B
1N967B	JANHCA1N967B	JANHCB1N967B	JANHCC1N967B	JANHCD1N967B	JANHCE1N967B
1N968B	JANHCA1N968B	JANHCB1N968B	JANHCC1N968B	JANHCD1N968B	JANHCE1N968B
1N969B	JANHCA1N969B	JANHCB1N969B	JANHCC1N969B	JANHCD1N969B	JANHCE1N969B
1N970B	JANHCA1N970B	JANHCB1N970B	JANHCC1N970B	JANHCD1N970B	JANHCE1N970B
1N971B	JANHCA1N971B	JANHCB1N971B	JANHCC1N971B	JANHCD1N971B	JANHCE1N971B
1N972B	JANHCA1N972B	JANHCB1N972B	JANHCC1N972B	JANHCD1N972B	JANHCE1N972B
1N973B	JANHCA1N973B	JANHCB1N973B	JANHCC1N973B	JANHCD1N973B	JANHCE1N973B
1N974B	JANHCA1N974B	JANHCB1N974B	JANHCC1N974B	JANHCD1N974B	JANHCE1N974B
1N975B	JANHCA1N975B	JANHCB1N975B	JANHCC1N975B	JANHCD1N975B	JANHCE1N975B
1N976B	JANHCA1N976B	JANHCB1N976B	JANHCC1N976B	JANHCD1N976B	JANHCE1N976B
1N977B	JANHCA1N977B	JANHCB1N977B	JANHCC1N977B	JANHCD1N977B	JANHCE1N977B
1N978B	JANHCA1N978B	JANHCB1N978B	JANHCC1N978B	JANHCD1N978B	JANHCE1N978B
1N979B	JANHCA1N979B	JANHCB1N979B	JANHCC1N979B	JANHCD1N979B	JANHCE1N979B
1N980B	JANHCA1N980B	JANHCB1N980B	JANHCC1N980B	JANHCD1N980B	JANHCE1N980B
1N981B	JANHCA1N981B	JANHCB1N981B	JANHCC1N981B	JANHCD1N981B	JANHCE1N981B
1N982B	JANHCA1N982B	JANHCB1N982B	JANHCC1N982B	JANHCD1N982B	JANHCE1N982B
1N983B	JANHCA1N983B	JANHCB1N983B	JANHCC1N983B	JANHCD1N983B	JANHCE1N983B
1N984B	JANHCA1N984B	JANHCB1N984B	JANHCC1N984B	JANHCD1N984B	JANHCE1N984B
1N985B	JANHCA1N985B	JANHCB1N985B	JANHCC1N985B	JANHCD1N985B	JANHCE1N985B
1N986B	JANHCA1N986B	JANHCB1N986B	JANHCC1N986B	JANHCD1N986B	JANHCE1N986B
1N987B	JANHCA1N987B	JANHCB1N987B	JANHCC1N987B	JANHCD1N987B	JANHCE1N987B
1N988B	JANHCA1N988B	JANHCB1N988B	JANHCC1N988B	JANHCD1N988B	JANHCE1N988B
1N989B	JANHCA1N989B	JANHCB1N989B	JANHCC1N989B	JANHCD1N989B	JANHCE1N989B
1N990B	JANHCA1N990B	JANHCB1N990B	JANHCC1N990B	JANHCD1N990B	JANHCE1N990B
1N991B	JANHCA1N991B	JANHCB1N991B	JANHCC1N991B	JANHCD1N991B	JANHCE1N991B
1N992B	JANHCA1N992B	JANHCB1N992B	JANHCC1N992B	JANHCD1N992B	JANHCE1N992B

(1) For JANKC level, replace JANHC with JANKC.
(2) PINs are shown using the "B" first suffix symbol. The PIN is also available with a "C" or "D" first suffix symbol, indicating the voltage tolerance.

6.8 Maximum power versus lead temperature. Typical maximum power rating as a function of lead temperature for various lead lengths is shown on figure 21.

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Inches	Millimeters
.000	0.00
.125	3.18
.250	6.35
.375	9.53

FIGURE 21. Maximum power versus lead temperature and lead length.

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6.9 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
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Navy − SH
Air Force − 85
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Army − AR, AV, MI, SM
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